PRACTICAL NO 1

Student Retention Analysis using Excel (Charts, KPIs and Dashboard)

Aim

To analyse student retention data in Microsoft Excel and create KPIs, charts and a simple dashboard for identifying students at risk.

Software / Tools Used

- Microsoft Excel 2019 / 2021 / 2024 / Microsoft 365 desktop

Objectives

- Create and format a student-retention dataset.
- Calculate retention KPIs using Excel formulas.
- Create charts for retention status and attendance.
- Assemble a one-sheet dashboard for quick decision-making.

Theory

Student retention analysis uses academic indicators such as attendance, internal marks and assignment completion to identify students who may need support. A KPI (Key Performance Indicator) is a measurable value used to monitor performance. In this practical, Retention Rate and At-Risk Count are key KPIs.

Dataset

Student_ID	Name	Attendance_%	Internal_Marks	Assignment_Submitted	Retention Status
1	Kabir	82	31	2	Retained
2	Rohan	58	15	1	At Risk
3	Sara	91	18	1	At Risk
4	Ishaan	67	35	3	Retained
5	Vaan	76	17	1	At Risk
6	Kiara	49	28	2	Retained
7	Aarav	88	20	1	At Risk
8	Diya	72	25	1	Retained
9	Priya	61	36	2	At Risk
10	Sneha	84	22	3	At Risk

Procedure

Step 1: Enter and Format the Data

Enter the dataset in cells **A1:F8**.

Select the range and press **Ctrl + T** to convert it into an Excel Table.

Keep **"My table has headers"** selected.

Expected Output / Observation:

The specified KPI, chart or dashboard element should appear and update when the source data changes.

N14

	A	B	C	D	E	F	G
1	Student_ID	Name	Attendance_%	Internal_Marks	Assignments_Subm	Retention Status	
2	1	Kabir	82	31	2	Retained	
3	2	Rohan	58	15	1	At Risk	
4	3	Sara	91	18	1	At Risk	
5	4	Ishaan	67	35	3	Retained	
6	5	Vaan	76	17	1	At Risk	
7	6	Kiara	49	28	2	Retained	
8	7	Aarav	88	20	1	At Risk	
9	8	Diya	72	25	1	Retained	
10	9	Priya	61	36	2	At Risk	
11	10	Sneha	84	22	3	At Risk	
12							

Step 2: Calculate Total Students

In a KPI cell, enter:

=COUNTA(Table1[Student_ID])

Expected Output / Observation:

The Total Students KPI should display the total number of students and update when the source data changes.

N14

	A	B	C	D	E	F	G	H	I
1	Student_ID	Name	Attendance_%	Internal_Marks	Assignments_Subm	Retention Status			
2	1	Kabir	82	31	2	Retained		Total Student	10
3	2	Rohan	58	15	1	At Risk			
4	3	Sara	91	18	1	At Risk			
5	4	Ishaan	67	35	3	Retained			
6	5	Vaan	76	17	1	At Risk			
7	6	Kiara	49	28	2	Retained			
8	7	Aarav	88	20	1	At Risk			
9	8	Diya	72	25	1	Retained			
10	9	Priya	61	36	2	At Risk			
11	10	Sneha	84	22	3	At Risk			
12									
13									

Step 3: Calculate Retained Students

Enter:

=COUNTIF(Table1[Retention_Status],”Retained")

Expected Output / Observation:

The Retained Students KPI should display the number of retained students and update when the source data changes.

M12

	A	B	C	D	E	F	G	H	I
1	Student_ID	Name	Attendance_%	Internal_Marks	Assignments_Subm	Retention Status			
2	1	Kabir	82	31	2	Retained		Total Student	10
3	2	Rohan	58	15	1	At Risk		Retained Student	4
4	3	Sara	91	18	1	At Risk			
5	4	Ishaan	67	35	3	Retained			
6	5	Vaan	76	17	1	At Risk			
7	6	Kiara	49	28	2	Retained			
8	7	Aarav	88	20	1	At Risk			
9	8	Diya	72	25	1	Retained			
10	9	Priya	61	36	2	At Risk			
11	10	Sneha	84	22	3	At Risk			
12									

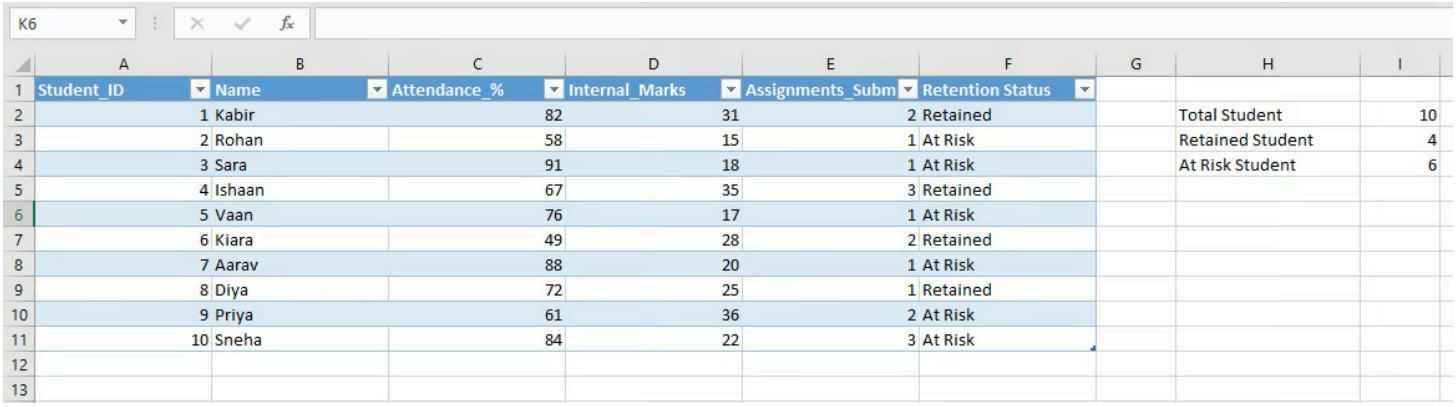
Step 4: Calculate At-Risk Students

Enter:

=COUNTIF(Table1[Retention_Status],"At Risk”)

Expected Output / Observation:

The At-Risk Students KPI should display the number of students at risk and update when the source data changes.



	A	B	C	D	E	F	G	H	I
1	Student_ID	Name	Attendance_%	Internal_Marks	Assignments_Subm	Retention Status			
2	1	Kabir	82	31	2	Retained		Total Student	10
3	2	Rohan	58	15	1	At Risk		Retained Student	4
4	3	Sara	91	18	1	At Risk		At Risk Student	6
5	4	Ishaan	67	35	3	Retained			
6	5	Vaan	76	17	1	At Risk			
7	6	Kiara	49	28	2	Retained			
8	7	Aarav	88	20	1	At Risk			
9	8	Diya	72	25	1	Retained			
10	9	Priya	61	36	2	At Risk			
11	10	Sneha	84	22	3	At Risk			
12									
13									

Step 5: Calculate Retention Rate

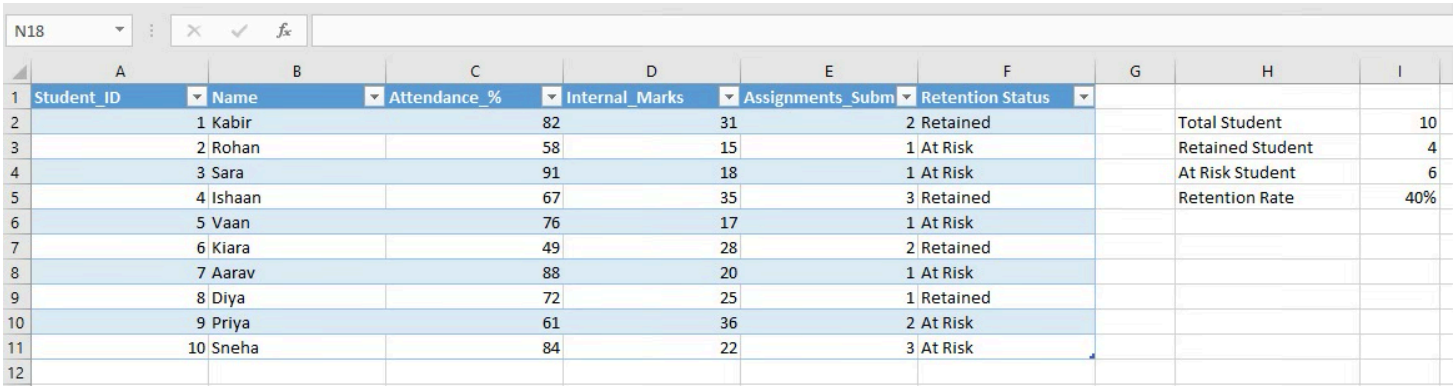
Enter:

=Retained_Students/Total_Students

Format the result as **Percentage**.

Expected Output / Observation:

The Retention Rate KPI should display the percentage of retained students and update when the source data changes.



	A	B	C	D	E	F	G	H	I
1	Student_ID	Name	Attendance_%	Internal_Marks	Assignments_Subm	Retention Status			
2	1	Kabir	82	31	2	Retained		Total Student	10
3	2	Rohan	58	15	1	At Risk		Retained Student	4
4	3	Sara	91	18	1	At Risk		At Risk Student	6
5	4	Ishaan	67	35	3	Retained		Retention Rate	40%
6	5	Vaan	76	17	1	At Risk			
7	6	Kiara	49	28	2	Retained			
8	7	Aarav	88	20	1	At Risk			
9	8	Diya	72	25	1	Retained			
10	9	Priya	61	36	2	At Risk			
11	10	Sneha	84	22	3	At Risk			
12									

Step 6: Calculate Average Attendance

Enter:

=AVERAGE(Table1[Attendance_%])

Expected Output / Observation:

The Average Attendance KPI should display the average attendance percentage and update when the source data changes.

Student_ID	Name	Attendance_%	Internal_Marks	Assignments_Subm	Retention Status
1	Kabir	82	31	2	Retained
2	Rohan	58	15	1	At Risk
3	Sara	91	18	1	At Risk
4	Ishaan	67	35	3	Retained
5	Vaan	76	17	1	At Risk
6	Kiara	49	28	2	Retained
7	Aarav	88	20	1	At Risk
8	Diya	72	25	1	Retained
9	Priya	61	36	2	At Risk
10	Sneha	84	22	3	At Risk

Total Student	10
Retained Student	4
At Risk Student	6
Retention Rate	40%
Average Attendance	72.8

Step 7: Create Retention Status Summary

Create a small two-row table containing **Retained** and **At Risk**, with COUNTIF formulas for each count.

Example:

Retention

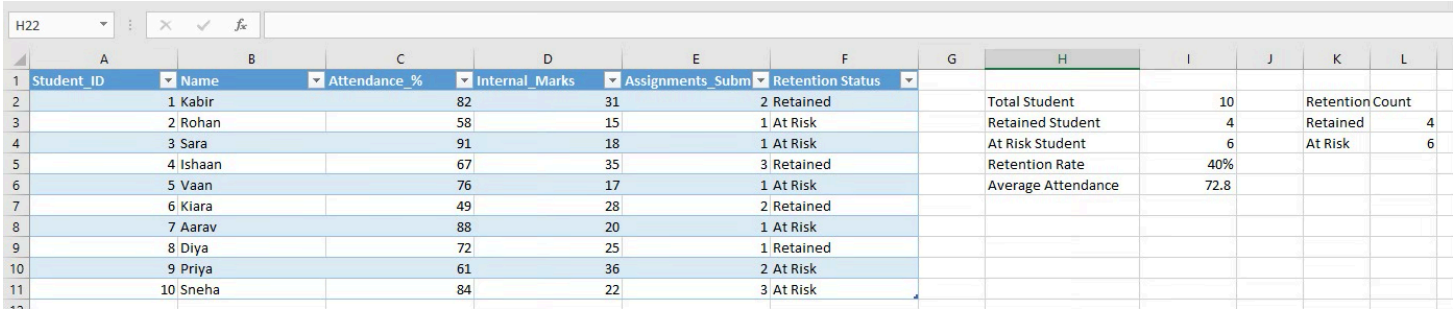
Status Count

Retained =COUNTIF(Table1[Retention_Status],"Retained")

At Risk =COUNTIF(Table1[Retention_Status],"At Risk")

Expected Output / Observation:

The summary table should display the count of students in each retention category and update when the source data changes



Student_ID	Name	Attendance_%	Internal_Marks	Assignments_Subm	Retention Status
1	Kabir	82	31	2	Retained
2	Rohan	58	15	1	At Risk
3	Sara	91	18	1	At Risk
4	Ishaan	67	35	3	Retained
5	Vaan	76	17	1	At Risk
6	Kiara	49	28	2	Retained
7	Aarav	88	20	1	At Risk
8	Diya	72	25	1	Retained
9	Priya	61	36	2	At Risk
10	Sneha	84	22	3	At Risk

Total Student	10	Retention	Count
Retained Student	4	Retained	4
At Risk Student	6	At Risk	6
Retention Rate	40%		
Average Attendance	72.8		

Step 8: Create Retention Chart

Select the summary table.

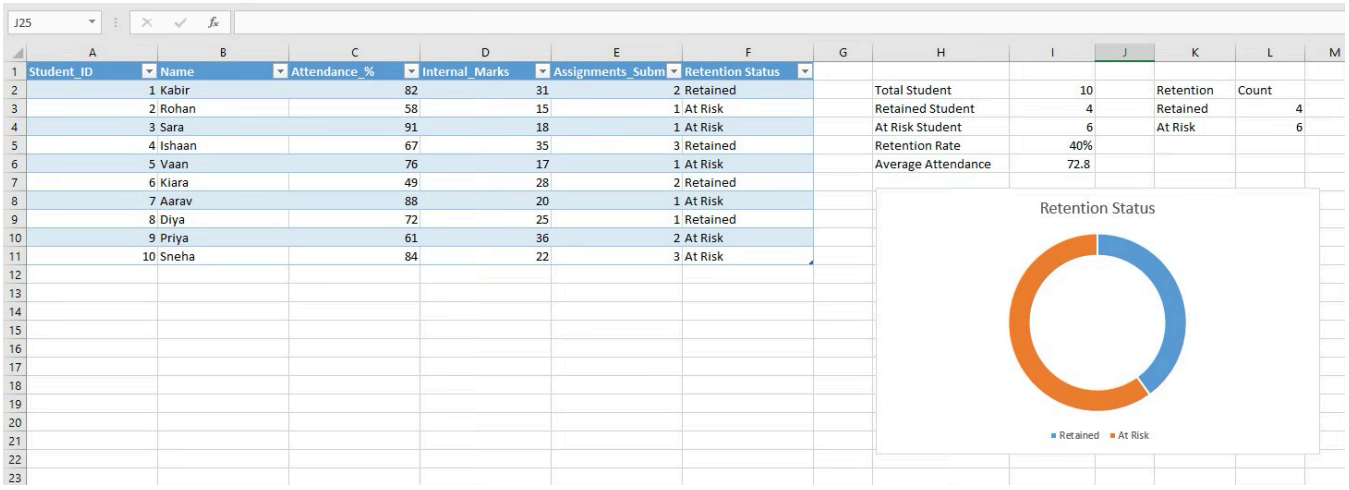
Go to:

Insert → Pie or Doughnut Chart → Doughnut

Add **Data Labels** and a clear chart title.

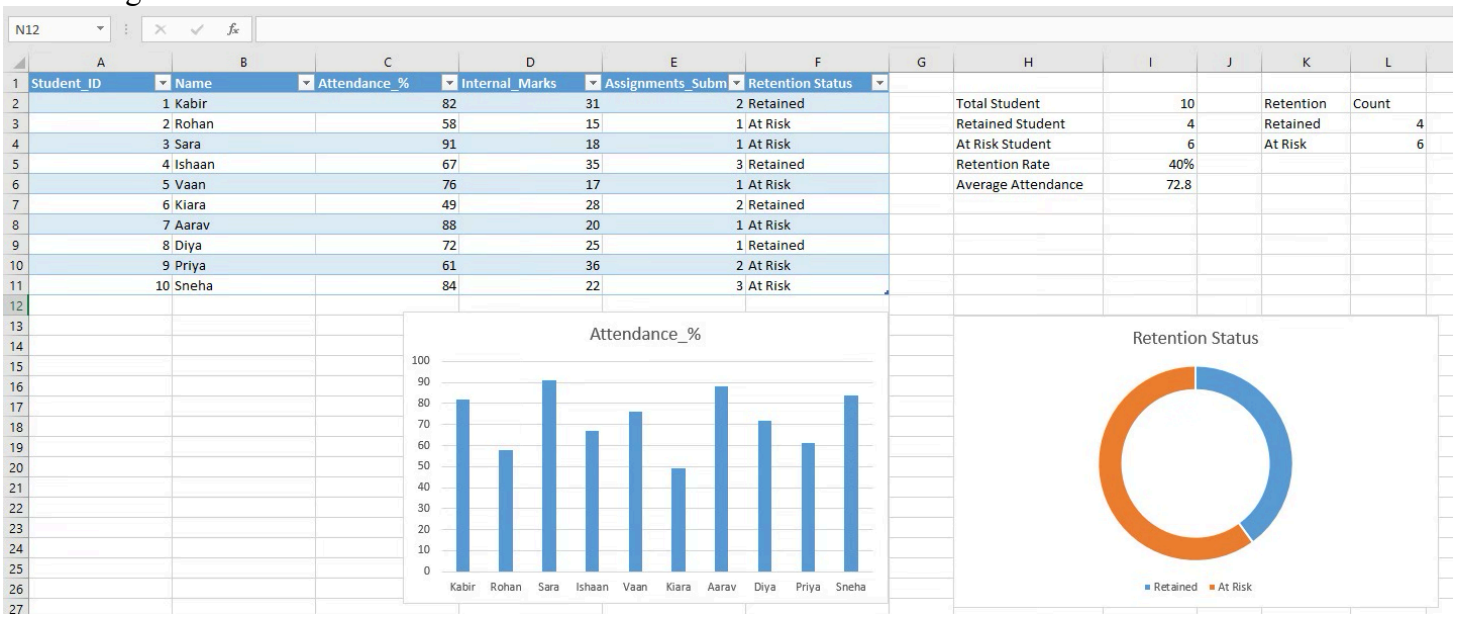
Expected Output / Observation:

A doughnut chart showing the distribution of **Retained** and **At Risk** students should appear and update when the source data changes.



Step 9: Create Attendance Chart
Select the **Name** and **Attendance_%** columns.
Go to:
Insert → Column or Bar Chart → Clustered Column
Give it the title:
Student Attendance Percentage

Expected Output / Observation:
A column chart showing the attendance percentage of each student should appear and update when the source data changes.

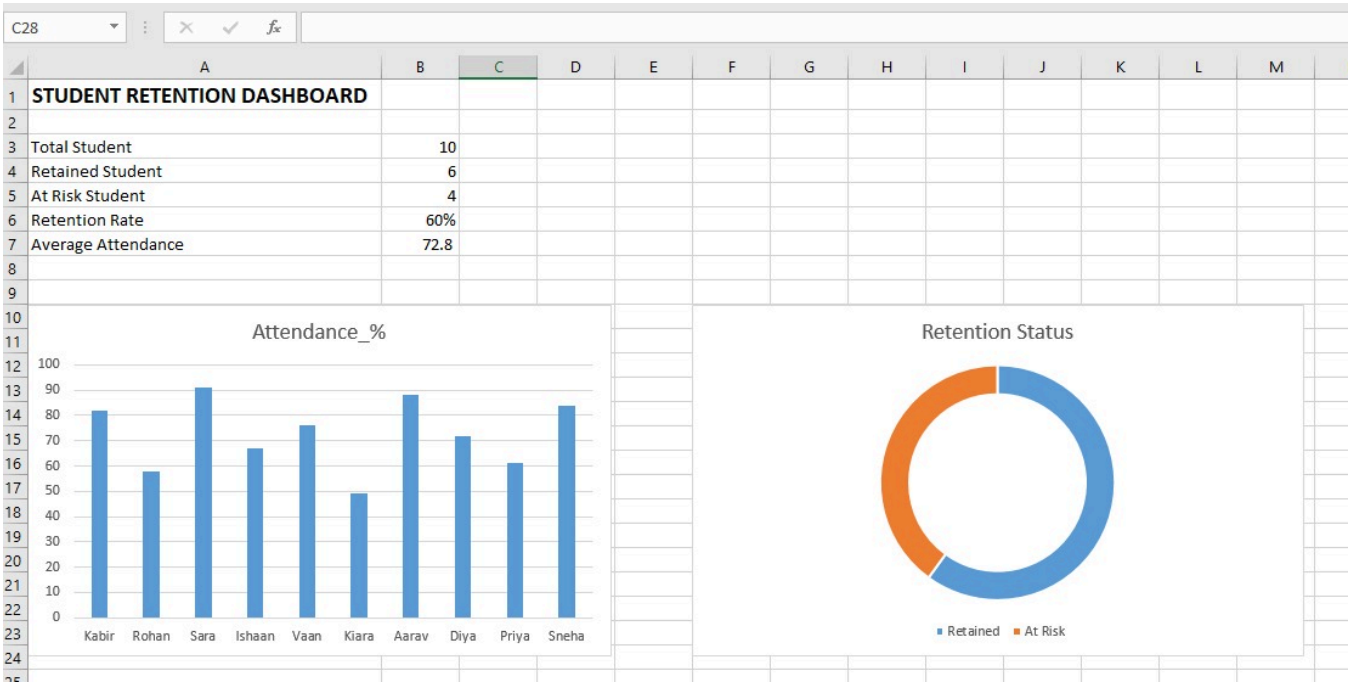


Student_ID	Name	Attendance_%	Internal_Marks	Assignments_Subm	Retention Status
1	Kabir	82	31	2	Retained
2	Rohan	58	15	1	At Risk
3	Sara	91	18	1	At Risk
4	Ishaan	67	35	3	Retained
5	Vaan	76	17	1	At Risk
6	Kiara	49	28	2	Retained
7	Aarav	88	20	1	At Risk
8	Diya	72	25	1	Retained
9	Priya	61	36	2	At Risk
10	Sneha	84	22	3	At Risk

Total Student	10	Retention	Count
Retained Student	4	Retained	4
At Risk Student	6	At Risk	6
Retention Rate	40%		
Average Attendance	72.8		

Step 10: Build the Dashboard
Create a new worksheet named **Dashboard**.
Place the KPI cells at the top and copy/move the two charts below.
Add a title such as:
Student Retention Dashboard

Expected Output / Observation:
A one-sheet dashboard containing the KPIs and supporting charts should be created and update when the source data changes.



Total Student	10
Retained Student	6
At Risk Student	4
Retention Rate	60%
Average Attendance	72.8

Step 11: Add Conditional Formatting

On the source sheet, select **Attendance_%**.

Go to:

Home → Conditional Formatting → Highlight Cells Rules → Less Than

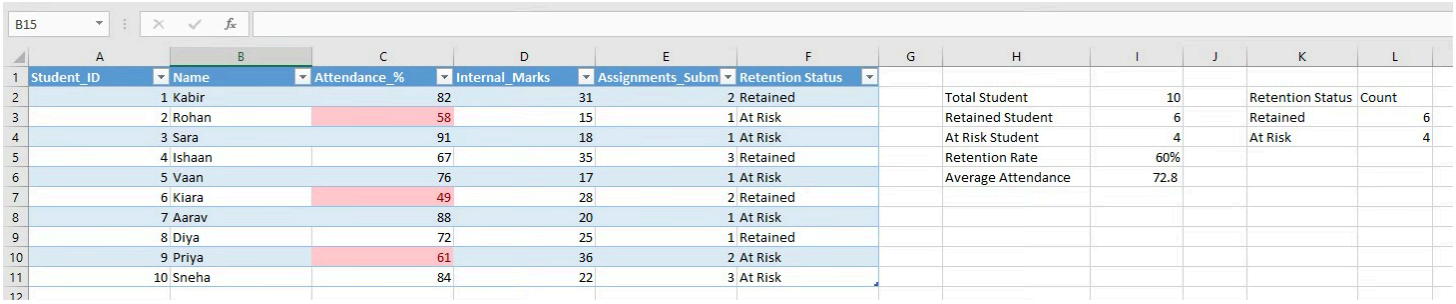
Enter:

65

This will highlight students having low attendance.

Expected Output / Observation:

Attendance values below 65 should be highlighted.



	A	B	C	D	E	F	G	H	I	J	K	L
1	Student_ID	Name	Attendance_%	Internal_Marks	Assignments_Subm	Retention Status						
2	1	Kabir	82	31	2	Retained		Total Student	10		Retention Status	Count
3	2	Rohan	58	15	1	At Risk		Retained Student	6		Retained	6
4	3	Sara	91	18	1	At Risk		At Risk Student	4		At Risk	4
5	4	Ishaan	67	35	3	Retained		Retention Rate	60%			
6	5	Vaan	76	17	1	At Risk		Average Attendance	72.8			
7	6	Kiara	49	28	2	Retained						
8	7	Aarav	88	20	1	At Risk						
9	8	Diya	72	25	1	Retained						
10	9	Priya	61	36	2	At Risk						
11	10	Sneha	84	22	3	At Risk						

Step 12: Interpret

Identify which students need intervention using the **At Risk** label together with low attendance and internal marks.

Result

A student retention dashboard with total students, retained count, at-risk count, retention rate, average attendance and supporting charts was created successfully.

Conclusion

Excel can combine formulas, conditional formatting and visualizations to convert student records into actionable retention information.

PRACTICAL NO 2
Data Exploration and Visualization (Types of Data and Summary Statistics)

Aim: To explore a dataset, identify data types, calculate summary statistics and create suitable visualizations in Excel.

Software / Tools Used
Microsoft Excel desktop

Objectives
- Identify categorical and numerical variables.
- Calculate mean, median, mode, minimum, maximum and standard deviation.
- Create a histogram and scatter plot.
- Interpret basic patterns and relationships.

Theory : Data exploration is the initial examination of a dataset to understand its structure, distribution and relationships. Summary statistics describe central tendency and dispersion, while charts help reveal trends, distributions and associations.

Dataset :

Student_ID	Department	Study_Hours	Attendance_%	Exam_Score
1	IT	2.5	68	55
2	CS	4.0	82	71
3	IT	5.5	90	84
4	DS	3.0	73	63
5	CS	6.0	95	88
6	IT	1.5	61	48
7	DS	4.5	85	78
8	CS	3.5	79	69
9	IT	5.0	88	81
10	DS	2.0	66	69

Step 1: Identify Data Types

Column	Type	Reason
Student_Id	Categorical identifier	Used as a label, not for arithmetic meaning
Department	Nominal Categorical	Categories have no natural order
Study_Hours	Numerical Continuous	Can take decimal values
Attendance_%	Numerical Continous	Measured percentage
Exam_Score	Numerical Quantitative	Measured score

Step: Mean Exam Score

Enter the following formula in a separate summary area:

Student_	Department	Study_Hou	Attendance_	Exam_Sco	mean exam score
1	IT	2.5	68	55	41.83333333
2	CS	4	82	71	52.33333333
3	IT	5.5	90	84	59.83333333
4	DS	3	73	63	46.33333333
5	CS	6	95	88	63
6	IT	1.5	61	48	36.83333333
7	DS	4.5	85	78	55.83333333
8	CS	3.5	79	69	50.5
9	IT	5	88	81	58
10	DS	2	66	69	45.66666667

Step: Median Exam Score

Enter the following formula in a separate summary area:

Student_	Department	Study_Hou	Attendance_	Exam_Sco	mean exam score	meadian exam score
1	IT	2.5	68	55	41.83333333	55
2	CS	4	82	71	52.33333333	71
3	IT	5.5	90	84	59.83333333	84
4	DS	3	73	63	46.33333333	63
5	CS	6	95	88	63	88
6	IT	1.5	61	48	36.83333333	48
7	DS	4.5	85	78	55.83333333	78
8	CS	3.5	79	69	50.5	69
9	IT	5	88	81	58	81
10	DS	2	66	69	45.66666667	66

Step: Mode Exam Score

Enter the following formula in a separate summary area:

Student_	Department	Study_Hou	Attendance_	Exam_Sco	mean exam score	meadian exam score	mode exam
1	IT	2.5	68	55	41.83333333	55	69
2	CS	4	82	71	52.33333333	71	
3	IT	5.5	90	84	59.83333333	84	
4	DS	3	73	63	46.33333333	63	
5	CS	6	95	88	63	88	
6	IT	1.5	61	48	36.83333333	48	
7	DS	4.5	85	78	55.83333333	78	
8	CS	3.5	79	69	50.5	69	
9	IT	5	88	81	58	81	
10	DS	2	66	69	45.66666667	66	

Step: Minimum & Maximum

Enter the following formula in a separate summary area:

Student_	Department	Study_Hou	Attendance_	Exam_Sco	mean exam score	meadian exam score	mode exam	min	max
1	IT	2.5	68	55	41.83333333	55	69	48	88
2	CS	4	82	71	52.33333333	71			
3	IT	5.5	90	84	59.83333333	84			
4	DS	3	73	63	46.33333333	63			
5	CS	6	95	88	63	88			
6	IT	1.5	61	48	36.83333333	48			
7	DS	4.5	85	78	55.83333333	78			
8	CS	3.5	79	69	50.5	69			
9	IT	5	88	81	58	81			
10	DS	2	66	69	45.66666667	66			

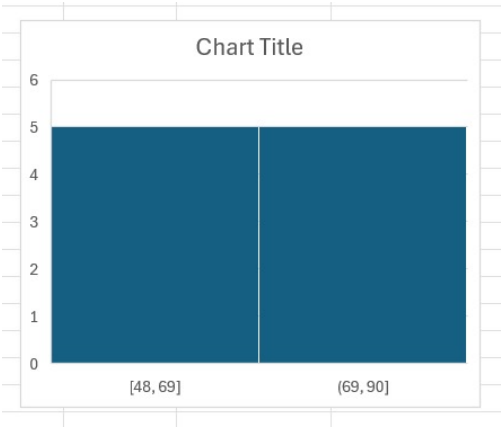
Step: Standard Deviation

Enter the following formula in a separate summary area:

Student_	Department	Study_Hou	Attendance_	Exam_Sco	mean exam score	meadian exam score	mode exam	min	max	standerd deviation
1	IT	2.5	68	55	41.83333333	55	69	48	88	12.74711296
2	CS	4	82	71	52.33333333	71				
3	IT	5.5	90	84	59.83333333	84				
4	DS	3	73	63	46.33333333	63				
5	CS	6	95	88	63	88				
6	IT	1.5	61	48	36.83333333	48				
7	DS	4.5	85	78	55.83333333	78				
8	CS	3.5	79	69	50.5	69				
9	IT	5	88	81	58	81				
10	DS	2	66	69	45.66666667	66				

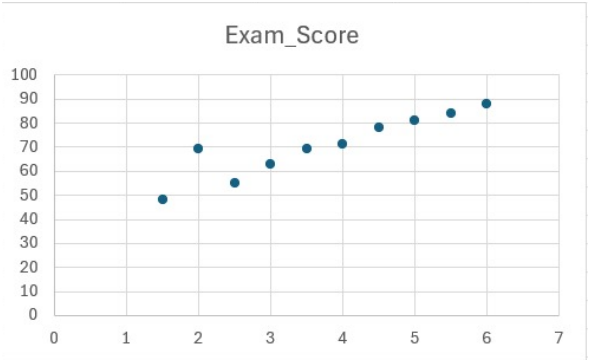
Step: Create a Histogram

Select Exam_Score values E2:E11. Go to Insert > Insert Statistic Chart > Histogram. Add the title Distribution of Exam Scores.



Step: Create a Scatter Plot

Select Study_Hours and Exam_Score columns. Go to Insert > Scatter (X, Y) > Scatter with only Markers. Use Study_Hours on the X-axis and Exam_Score on the Y-axis. Add chart and axis titles.



Result

Data types, summary statistics, a histogram and a scatter plot were generated successfully.

Conclusion

Exploratory analysis helps understand the dataset before advanced BI or predictive techniques are applied.

PRACTICAL NO 3
Executive Dashboard Design using Power BI

Aim : To import business sales data into Power BI Desktop and design an interactive executive dashboard using KPI cards, charts and slicers.

Software / Tools Used
Microsoft Power BI Desktop; Microsoft Excel for creating the source workbook

Objectives
- Import an Excel table into Power BI Desktop.
- Create basic DAX measures.
- Design KPI cards and business charts.
- Add slicers for interactive filtering.
- Format and save an executive report.

Theory : An executive dashboard presents high-level KPIs and trends in a compact visual form. Power BI uses a semantic model, DAX measures and interactive visualizations so decision-makers can filter and explore results.

Source Dataset :

Date	Region	Category	Product	Sales	Profit	Quantity
2026-01-05	West	Electronics	Laptop	75000	12000	3
2026-01-12	East	Electronics	Phone	42000	7000	6
2026-02-02	West	Furniture	Chair	18000	3500	8
2026-02-15	North	Furniture	Desk	26000	4200	4
2026-03-01	South	Electronics	Tablet	33000	5500	5
2026-03-18	East	Office Supplies	Printer	22000	3000	3
2026-04-08	North	Electronics	Laptop	82000	14000	3
2026-04-22	South	Office Supplies	Paper	9000	1800	30
2026-05-10	West	Furniture	Desk	31000	5200	5
2026-05-25	East	Electronics	Phone	47000	7600	7
2026-06-11	North	Office Supplies	Printer	24000	3400	4
2026-06-28	South	Electronics	Laptop	78000	12500	3

Step 1: Prepare the Excel File
Enter the dataset in Excel, select it, press Ctrl+T to convert it into a Table, name the table SalesData from Table Design, and save the workbook as BI_Sales.xlsx.

Date	Region	Category	Product	Sales	Profit	Quantity
05-01-2026	West	Electronics	Laptop	75000	12000	3
12-02-2026	East	Electronics	Phone	42000	7000	6
02-02-2026	West	Furniture	Chair	18000	3500	8
15-02-2026	North	Furniture	Desk	26000	4200	4
01-03-2026	South	Electronics	Tablet	33000	5500	5
18-03-2026	East	Office Suppl	Printer	22000	3000	3
08-04-2026	North	Electronics	Laptop	82000	14000	3
22-04-2026	South	Office Suppl	Paper	9000	1800	30
10-05-2026	West	Furniture	Desk	31000	5200	5
25-05-2026	East	Electronics	Phone	47000	7600	7
11-06-2026	North	Office Suppl	Printer	24000	3400	4
28-06-2026	South	Electronics	Laptop	78000	12500	3

Step 2: Import into Power BI

Open Power BI Desktop. Select Home > Get data > Excel. Choose BI_Sales.xlsx, select SalesData in Navigator, and choose Load.

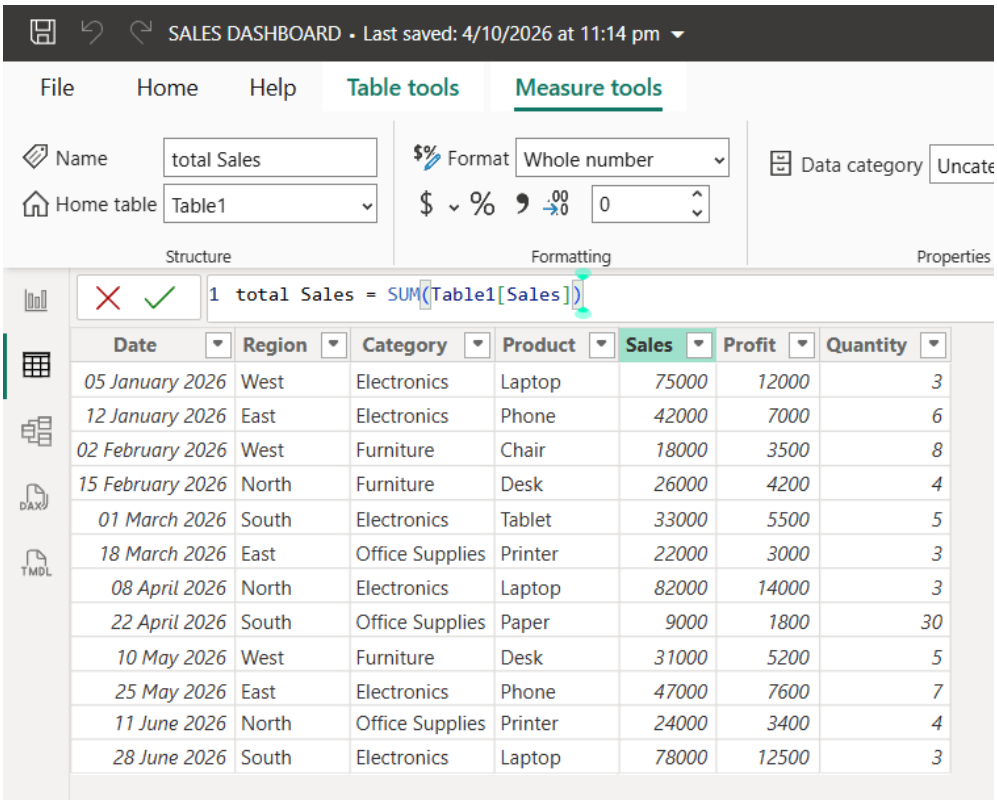
Date	Region	Category	Product	Sales	Profit	Quantity
05 January 2026	West	Electronics	Laptop	*75000*	*12000*	*3*
12 January 2026	East	Electronics	Phone	*42000*	*7000*	*6*
02 February 2026	West	Furniture	Chair	*18000*	*3500*	*8*
15 February 2026	North	Furniture	Desk	*26000*	*4200*	*4*
01 March 2026	South	Electronics	Tablet	*33000*	*5500*	*5*
18 March 2026	East	Office Supplies	Printer	*22000*	*3000*	*3*
08 April 2026	North	Electronics	Laptop	*82000*	*14000*	*3*
22 April 2026	South	Office Supplies	Paper	*9000*	*1800*	*30*
10 May 2026	West	Furniture	Desk	*31000*	*5200*	*5*
25 May 2026	East	Electronics	Phone	*47000*	*7600*	*7*
11 June 2026	North	Office Supplies	Printer	*24000*	*3400*	*4*
28 June 2026	South	Electronics	Laptop	*78000*	*12500*	*3*

Step 3: Check Data Types

Open Data view or Transform Data if needed. Ensure Date is Date type, Sales/Profit/Quantity are numeric, and Region/Category/Product are text. If you use Power Query, select Home > Close & Apply after corrections.

Step 4: Create Measures

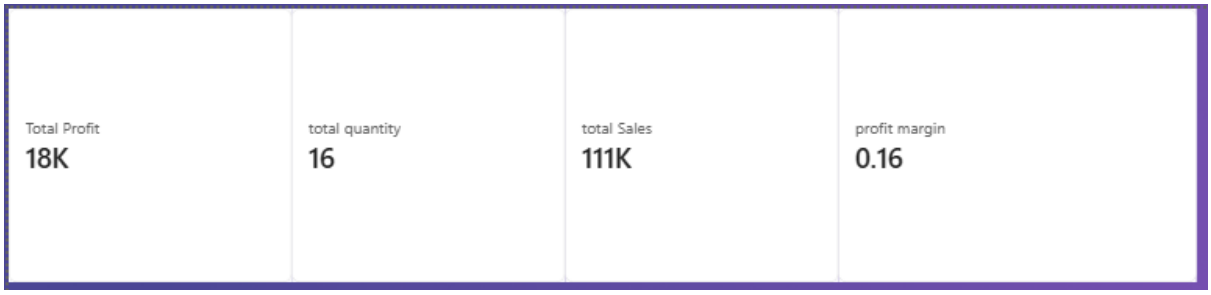
In the Data pane, right-click SalesData and choose New measure (or use Home > New measure). Create the following measures one by one:



Date	Region	Category	Product	Sales	Profit	Quantity
05 January 2026	West	Electronics	Laptop	75000	12000	3
12 January 2026	East	Electronics	Phone	42000	7000	6
02 February 2026	West	Furniture	Chair	18000	3500	8
15 February 2026	North	Furniture	Desk	26000	4200	4
01 March 2026	South	Electronics	Tablet	33000	5500	5
18 March 2026	East	Office Supplies	Printer	22000	3000	3
08 April 2026	North	Electronics	Laptop	82000	14000	3
22 April 2026	South	Office Supplies	Paper	9000	1800	30
10 May 2026	West	Furniture	Desk	31000	5200	5
25 May 2026	East	Electronics	Phone	47000	7600	7
11 June 2026	North	Office Supplies	Printer	24000	3400	4
28 June 2026	South	Electronics	Laptop	78000	12500	3

Step 5: Create KPI Cards

In Report view, choose the Card visual from the Visualizations gallery. Add Total Sales to Values. Create additional cards for Total Profit, Total Quantity and Profit Margin.



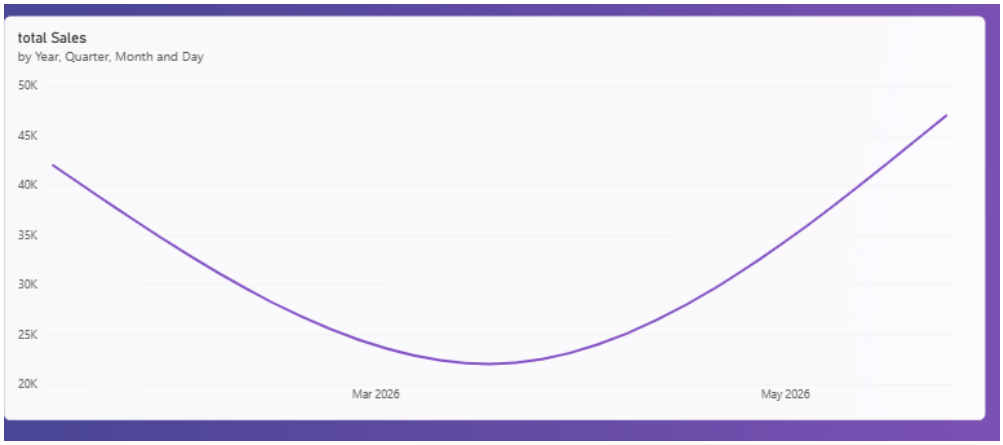
Step 6: Sales by Region

Insert a Clustered column chart. Place Region on the X-axis and Total Sales on the Y-axis/Values.



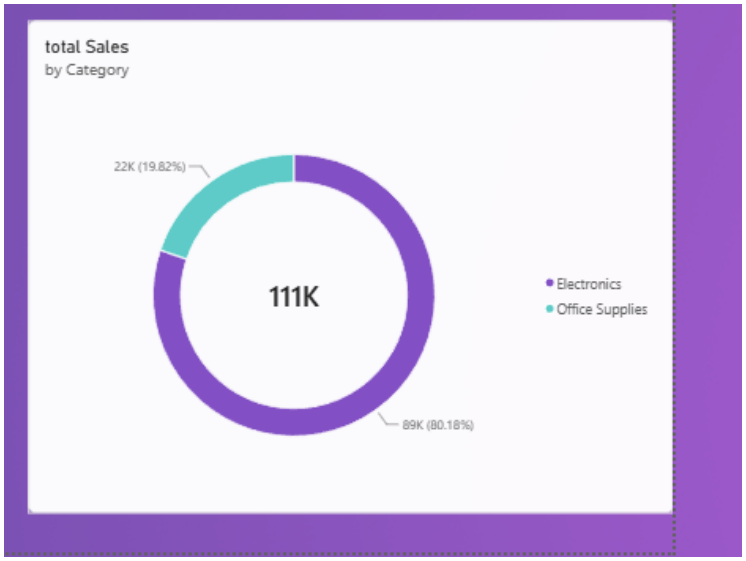
Step 7: Sales Trend

Insert a line chart. Place Date on the X-axis and Total Sales on the Y-axis. If Power BI creates a date hierarchy, use Month/Date as appropriate for the report.



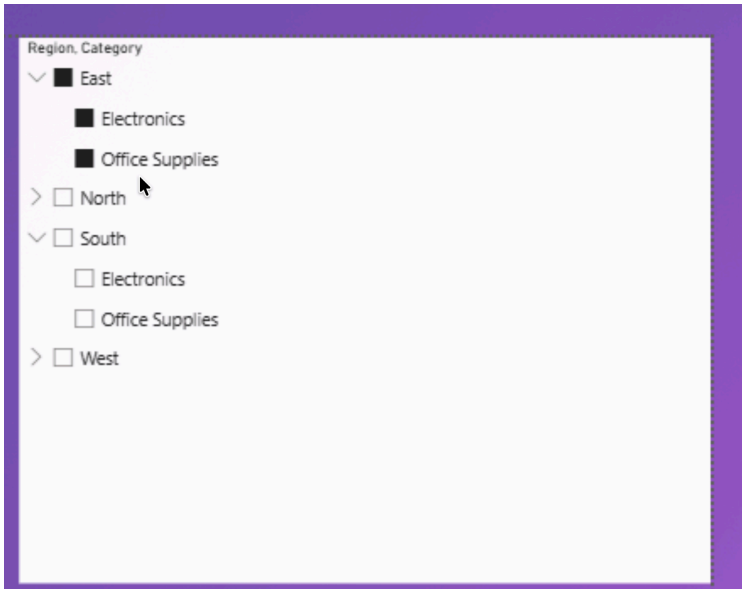
Step 8: Category Contribution

Insert a Donut chart. Place Category in Legend and Total Sales in Values.



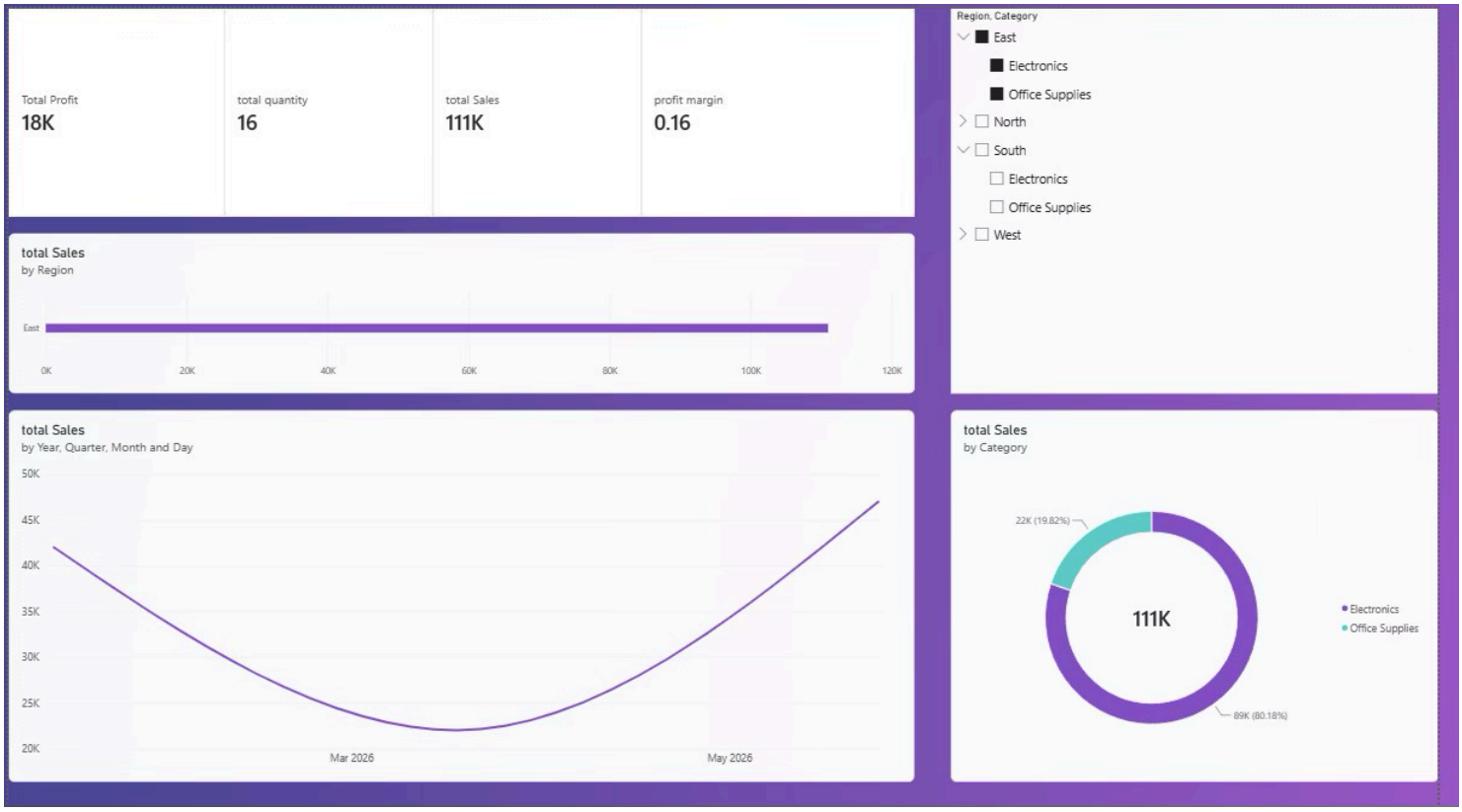
Step 9: Add Slicers

Insert Slicer visuals for Region and Category. Select items in the slicers and confirm that cards and charts update interactively.



Step 10: Format and Save

Add the page title Executive Sales Dashboard. Align visuals, format numbers using currency/commas as appropriate, and save the report as BI_Executive_Dashboard.pbix.



Result

An interactive executive sales dashboard was created in Power BI using measures, cards, charts and slicers.

Conclusion

Power BI integrates data modelling, DAX and interactive visualizations to support executive-level monitoring and decision-making.

PRACTICAL NO 4
Business Reporting and Visualization (Charts, Graphs and Dashboard)

Aim : To create a business sales report in Excel using PivotTables, PivotCharts and a compact dashboard.

Software / Tools Used
Microsoft Excel desktop

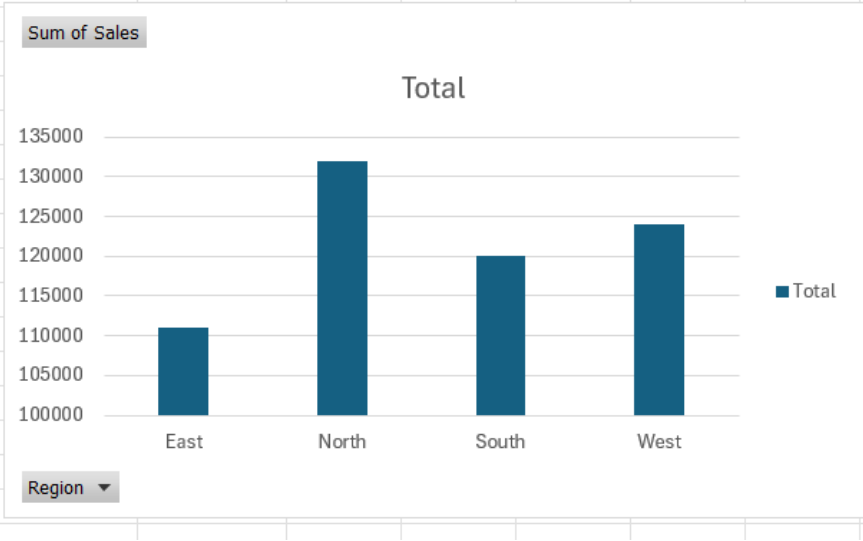
Objectives
- Create a PivotTable for business reporting.
- Summarize sales by region and category.
- Create multiple business charts.
- Use slicers to make a dashboard interactive.

Theory : Business reporting converts transaction data into structured summaries that managers can use. PivotTables aggregate large datasets quickly, while PivotCharts and slicers make the report visual and interactive. Use the same SalesData dataset from Practical 3 in Excel.

Step 1: Create a PivotTable
Click anywhere inside the SalesData table. Select Insert > PivotTable > From Table/Range > New Worksheet > OK.

Step 2: Region-wise Sales Report
Drag Region to Rows and Sales to Values. Ensure Sales is summarized by Sum.

Row Labels	Sum of Sales
East	111000
North	132000
South	120000
West	124000
Grand Total	**487000**

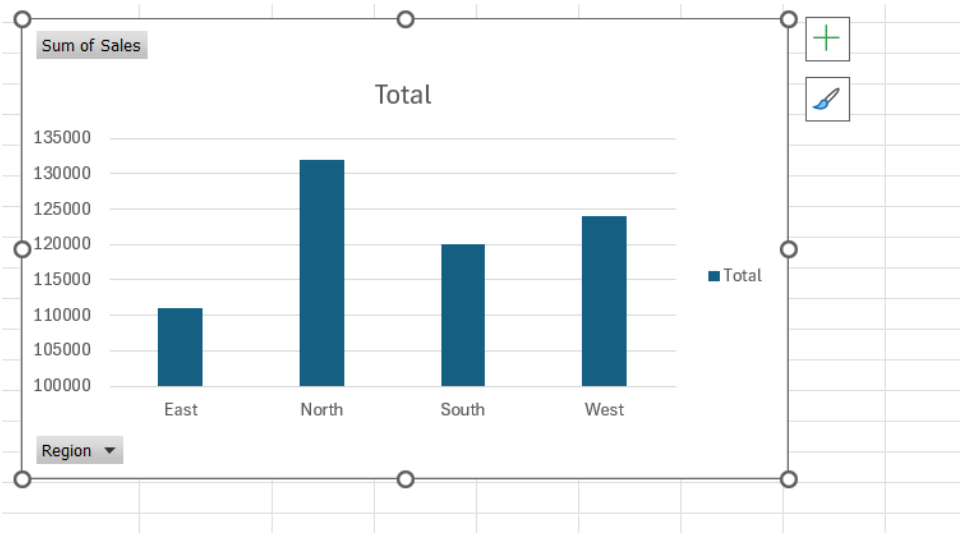


Step 3: Category-wise Profit Report
Create another PivotTable. Drag Category to Rows and Profit to Values.

Row Labels	Sum of Profit
Electronics	58600
Furniture	12900
Office Supplies	8200
Grand Total	**79700**

Step 4: Create a Column PivotChart

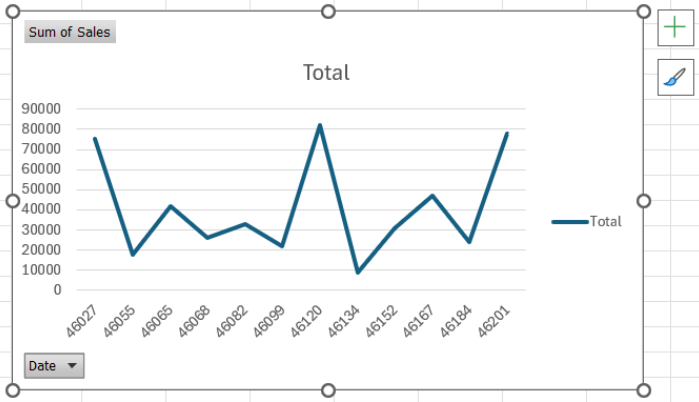
Select the region-wise PivotTable. Go to PivotTable Analyze > PivotChart and choose Clustered Column.



Step 5: Create a Line Chart for Trend

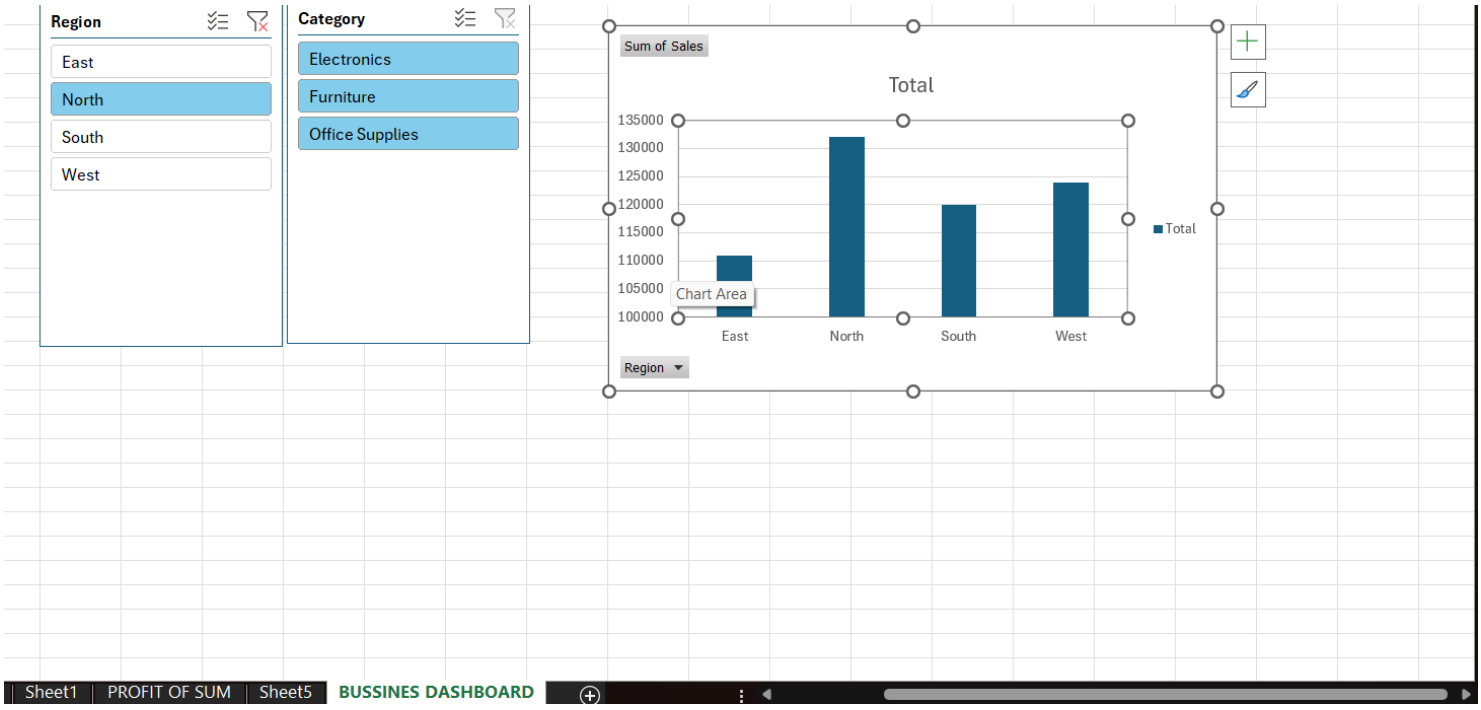
Create a PivotTable with Date in Rows and Sales in Values. Insert a Line PivotChart. Group dates by Months if required using right-click on a PivotTable date > Group > Months.

Row Labels	Sum of Sales
05 January 2026	75000
02 February 2026	18000
12 February 2026	42000
15 February 2026	26000
01 March 2026	33000
18 March 2026	22000
08 April 2026	82000
22 April 2026	9000
10 May 2026	31000
25 May 2026	47000
11 June 2026	24000
28 June 2026	78000
Grand Total	**487000**



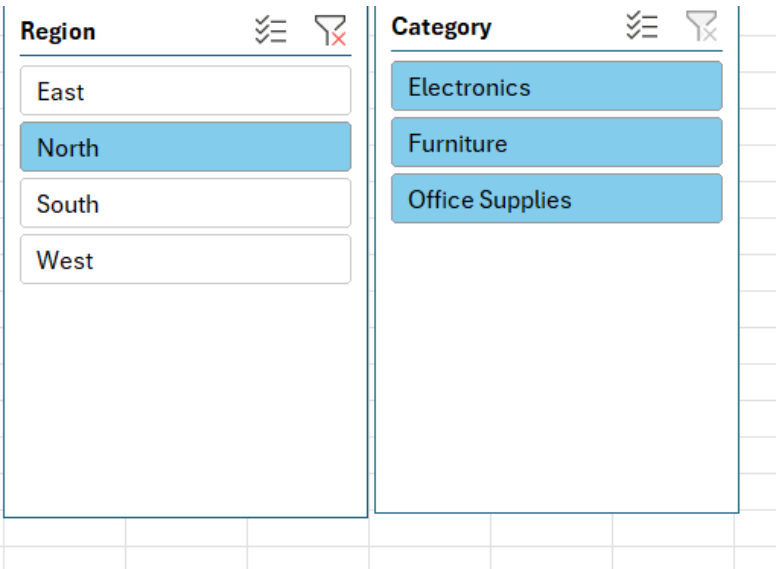
Step 6: Create a Dashboard Sheet

Insert a new sheet named Business Dashboard. Move or copy the PivotCharts to this sheet and arrange them neatly.



Step 7: Add Slicers

Click a PivotTable, choose PivotTable Analyze > Insert Slicer, select Region and Category, and place the slicers on the dashboard. Use Report Connections/PivotTable Connections to connect the slicer to multiple PivotTables if they use the same data source.



Result

A business report and interactive Excel dashboard were created using PivotTables, PivotCharts and slicers.

Conclusion

Pivot-based reporting is a fast way to summarize and visualize business data without writing formulas for every aggregation.

PRACTICAL NO 5
Data Warehousing Concepts: Star and Snowflake Schema Design

Aim : To study dimensional modelling concepts and design star and snowflake schemas for a sales data warehouse.

Software / Tools Used
Microsoft Word / PowerPoint / draw.io for drawing; no DBMS implementation is required for this model-study practical

Objectives
- Identify facts, dimensions and grain.
- Design a star schema.
- Convert a dimension into a snowflake structure.
- Compare star and snowflake schemas.

Theory : A data warehouse stores integrated historical data for analysis. Dimensional modelling commonly uses a central fact table containing measurable values and dimension tables containing descriptive attributes. A star schema connects dimensions directly to the fact table, while a snowflake schema normalizes one or more dimensions into related tables.

Scenario : A retail company wants to analyse sales by date, product, customer and store.

Step 1: Define the Grain
Grain means the level of detail represented by one row of the fact table. Define the grain as: one row per product sold in one transaction at one store on one date.

Step 2: Identify Measures
- Quantity
- Sales_Amount
- Discount
- Profit

Step 3: Identify Dimensions
- Date
- Product
- Customer
- Store

Step 4: Design the Star Schema

Table	Important Fields
Sales_Fact	Sales_Key (PK), Date_Key (FK), Product_Key (FK), Customer_Key (FK), Store_Key (FK), Quantity, Sales_Amount, Discount, Profit
Date_Dim	Date_Key (PK), Full_Date, Day, Month, Quarter, Year
Product_Dim	Product_Key (PK), Product_Name, Brand, Category
Customer_Dim	Customer_Key (PK), Customer_Name, Gender, City, Segment
Store_Dim	Store_Key (PK), Store_Name, City, Region

Step 5: Convert Product Dimension to Snowflake

Table	Fields
Product_Dim	Product_Key (PK), Product_Name, Brand, Category_Key (FK)
Category_Dim	Category_Key (PK), Category_Name, Department

Move Category-related attributes into a separate Category_Dim and keep Category_Key as a foreign key in Product_Dim.

Step 6: Compare the Schemas

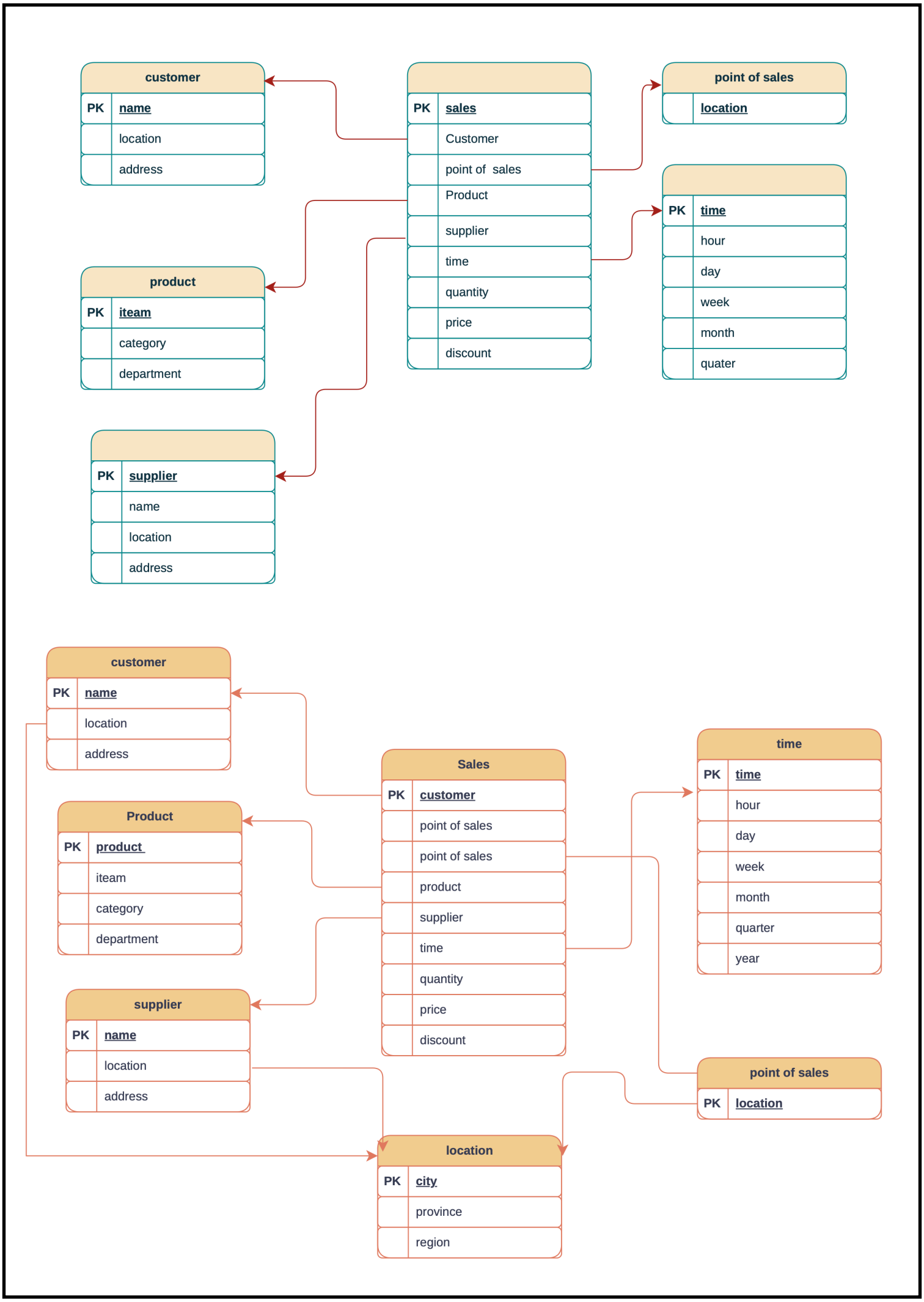
Star Schema	Snowflake Schema
Denormalized dimensions	Normalized dimensions
Simpler structure	More tables and joins
Usually easier for BI users	Can reduce. Redundancy
Often faster for simple analytical queries	More complex query paths

Result

Star and snowflake schemas for a retail sales data warehouse were designed successfully.

Conclusion

Dimensional modelling organizes warehouse data into facts and dimensions so that BI queries can analyse measures across business perspectives.


customer
PK name
location
address
PK sales
Customer
point of sales
Product
supplier
time
quantity
price
discount
point of sales
location
PK time
hour
day
week
month
quater
product
PK iteam
category
department
PK supplier
name
location
address
customer
PK name
location
address
Sales
PK customer
point of sales
point of sales
product
supplier
time
quantity
price
discount
time
PK time
hour
day
week
month
quarter
year
Product
PK product
iteam
category
department
supplier
PK name
location
address
point of sales
PK location
location
PK city
province
region

PRACTICAL NO 6
Data Preprocessing and Cleaning Techniques

Aim : To clean a small business dataset in Excel by handling duplicates, missing values, inconsistent text and outliers.

Software / Tools Used
Microsoft Excel desktop

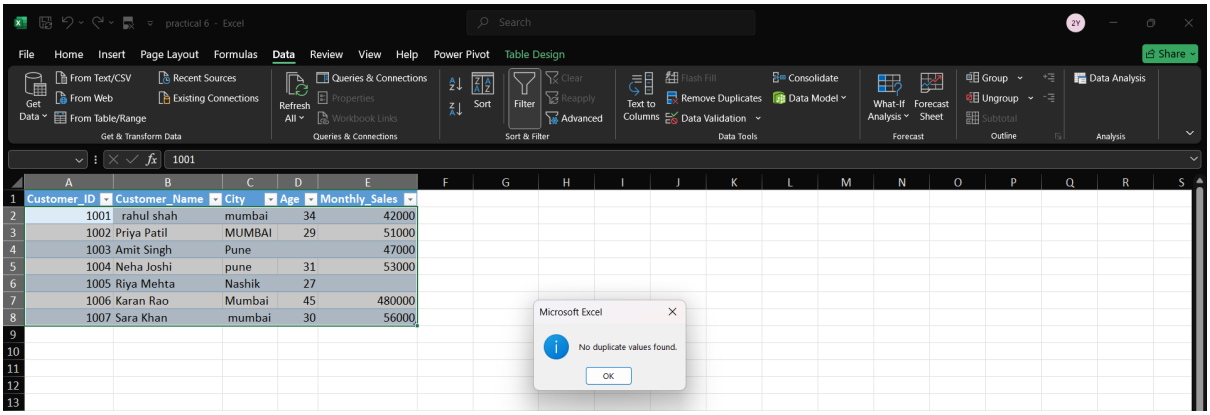
Objectives
- Identify common data-quality problems.
- Remove duplicates safely.
- Clean inconsistent text values.
- Handle missing numerical values.
- Detect possible outliers.

Theory : Data preprocessing improves data quality before analysis. Common issues include duplicate records, missing values, inconsistent labels, extra spaces, wrong data types and extreme values. Cleaning must be performed carefully because inappropriate corrections can distort the analysis.

Unclean Dataset

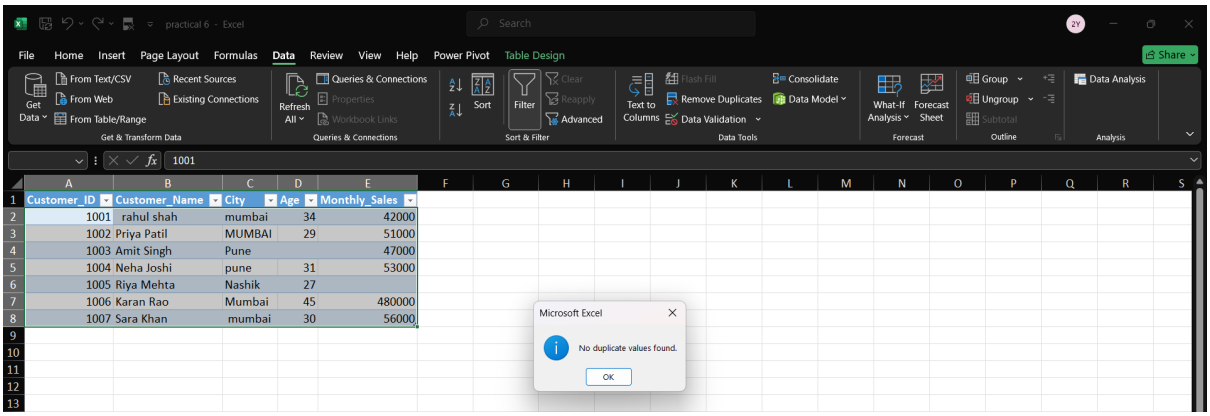
Customer_ID	Customer_Name	City	Age	Monthly_Sales
1001	rahul shah	mumbai	34	42000
1002	Priya Patil	MUMBAI	29	51000
1003	Amit Singh	Pune		47000
1004	Neha Joshi	pune	31	53000
1004	Neha Joshi	pune	31	53000
1005	Riya Mehta	Nashik	27	
1006	Karan Rao	Mumbai	45	480000
1007	Sara Khan	mumbai	30	56000

Step 1: Preserve the Raw Data
Make a copy of the original sheet and rename it Raw_Data. Perform all cleaning on a duplicate sheet named Clean_Data.



Step 2: Remove Duplicate Records

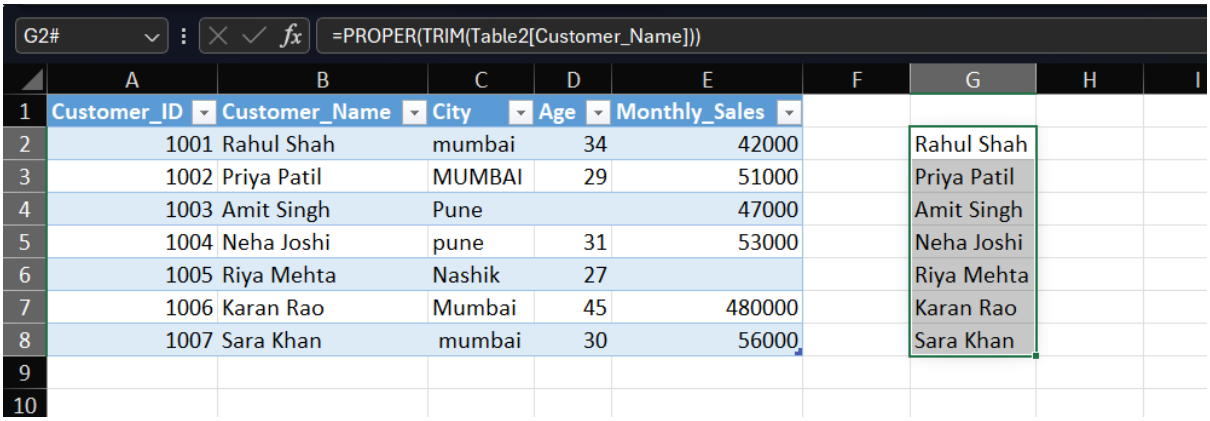
Select the full dataset on Clean_Data. Go to Data > Remove Duplicates. Keep all columns selected and click OK. The repeated Customer_ID 1004 row should be removed because the complete row is duplicated.



Customer_ID	Customer_Name	City	Age	Monthly_Sales
1001	rahul shah	mumbai	34	42000
1002	Priya Patil	MUMBAI	29	51000
1003	Amit Singh	Pune		47000
1004	Neha Joshi	pune	31	53000
1005	Riya Mehta	Nashik	27	
1006	Karan Rao	Mumbai	45	480000
1007	Sara Khan	mumbai	30	56000

Step 3: Clean Customer Names

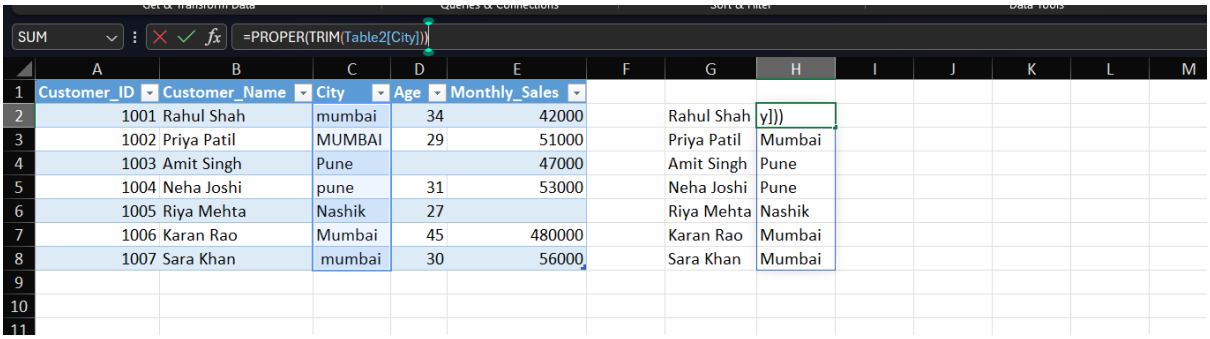
In a helper column enter =PROPER(TRIM(B2)) and fill down. TRIM removes extra spaces; PROPER converts names to title case. Copy the cleaned values and Paste Special > Values back if required.



Customer_ID	Customer_Name	City	Age	Monthly_Sales	
1001	Rahul Shah	mumbai	34	42000	Rahul Shah
1002	Priya Patil	MUMBAI	29	51000	Priya Patil
1003	Amit Singh	Pune		47000	Amit Singh
1004	Neha Joshi	pune	31	53000	Neha Joshi
1005	Riya Mehta	Nashik	27		Riya Mehta
1006	Karan Rao	Mumbai	45	480000	Karan Rao
1007	Sara Khan	mumbai	30	56000	Sara Khan

Step 4: Standardize City Names

Use =PROPER(TRIM(C2)) in a helper column and fill down.



Customer_ID	Customer_Name	City	Age	Monthly_Sales		
1001	Rahul Shah	mumbai	34	42000	Rahul Shah	y]))
1002	Priya Patil	MUMBAI	29	51000	Priya Patil	Mumbai
1003	Amit Singh	Pune		47000	Amit Singh	Pune
1004	Neha Joshi	pune	31	53000	Neha Joshi	Pune
1005	Riya Mehta	Nashik	27		Riya Mehta	Nashik
1006	Karan Rao	Mumbai	45	480000	Karan Rao	Mumbai
1007	Sara Khan	mumbai	30	56000	Sara Khan	Mumbai

Step 5: Handle Missing Age

Calculate the median age in a separate cell using =MEDIAN(D2:D8). Excel ignores blank cells. Enter the median value into the missing Age cell for Customer_ID 1003 and document that the value was imputed.

	Customer_ID	Customer_Name	City	Age	Monthly_Sales				missing age
2	1001	Rahul Shah	mumbai	34	42000		Rahul Shah	Mumbai	30.5
3	1002	Priya Patil	MUMBAI	29	51000		Priya Patil	Mumbai	
4	1003	Amit Singh	Pune	30.5	47000		Amit Singh	Pune	
5	1004	Neha Joshi	pune	31	53000		Neha Joshi	Pune	
6	1005	Riya Mehta	Nashik	27			Riya Mehta	Nashik	
7	1006	Karan Rao	Mumbai	45	480000		Karan Rao	Mumbai	
8	1007	Sara Khan	mumbai	30	56000		Sara Khan	Mumbai	
9									

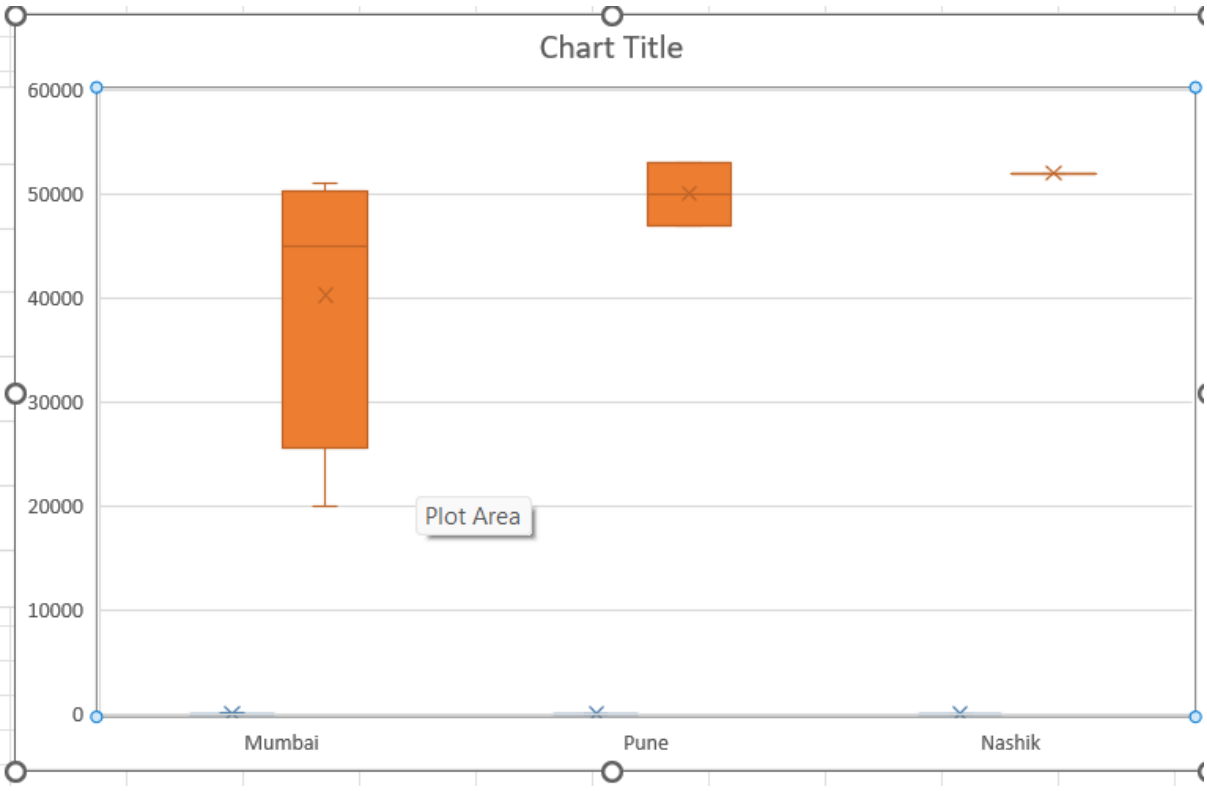
Step 6: Handle Missing Monthly Sales

Calculate the median of nonblank Monthly_Sales values using =MEDIAN(E2:E8), then replace the blank sales value with that median.

Customer_ID	Customer_Name	City	Age	Monthly_Sales				missing age	missing handle month sales
1001	Rahul Shah	mumbai	34	42000		Rahul Shah	Mumbai	30.5	52000
1002	Priya Patil	MUMBAI	29	51000		Priya Patil	Mumbai		
1003	Amit Singh	Pune	30.5	47000		Amit Singh	Pune		
1004	Neha Joshi	pune	31	53000		Neha Joshi	Pune		
1005	Riya Mehta	Nashik	27	52000		Riya Mehta	Nashik		
1006	Karan Rao	Mumbai	45	480000	(Ctrl)	Karan Rao	Mumbai		
1007	Sara Khan	mumbai	30	56000		Sara Khan	Mumbai		

Step 7: Detect Possible Outlier

Select Monthly_Sales and choose Home > Conditional Formatting > Top/Bottom Rules > Top 10 Items; change 10 to 1, or use a Box & Whisker chart from Insert > Statistic Chart. The value 480000 should be investigated as a possible outlier rather than deleted automatically.



Step 8: Final Quality Check

Verify that Customer_ID values are unique, names/cities are standardized, no required cells are blank and numeric columns contain numeric values.

Result

Duplicate, missing, inconsistent and extreme values were identified and cleaned in Excel.

PRACTICAL NO 7
Regression Analysis for Business Decision Making

Aim : To perform simple linear regression in Excel and interpret the relationship between advertising expenditure and sales.

Software / Tools Used

Microsoft Excel desktop with Analysis ToolPak enabled

Objectives

- Enable and use the Analysis ToolPak.
- Run a simple regression model.
- Interpret R Square, coefficient and p-value.
- Use the regression relationship for business prediction.

Theory : Regression estimates the relationship between a dependent variable and one or more independent variables. In simple linear regression, Y = a + bX, where Y is the outcome, X is the predictor, a is the intercept and b is the slope.

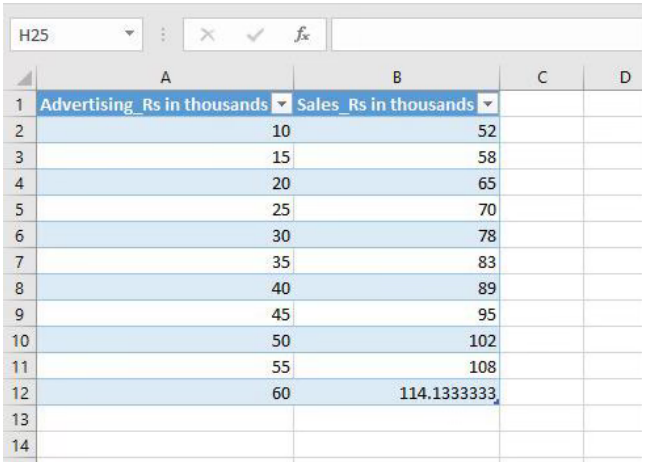
Dataset

This is Regression between **advertising expenditure and sales**

Advertising_Rs in thousands	Sales_Rs in thousands
10	52
15	58
20	65
25	70
30	78
35	83
40	89
45	95
50	102
55	108

Step 1: Enable Analysis ToolPak

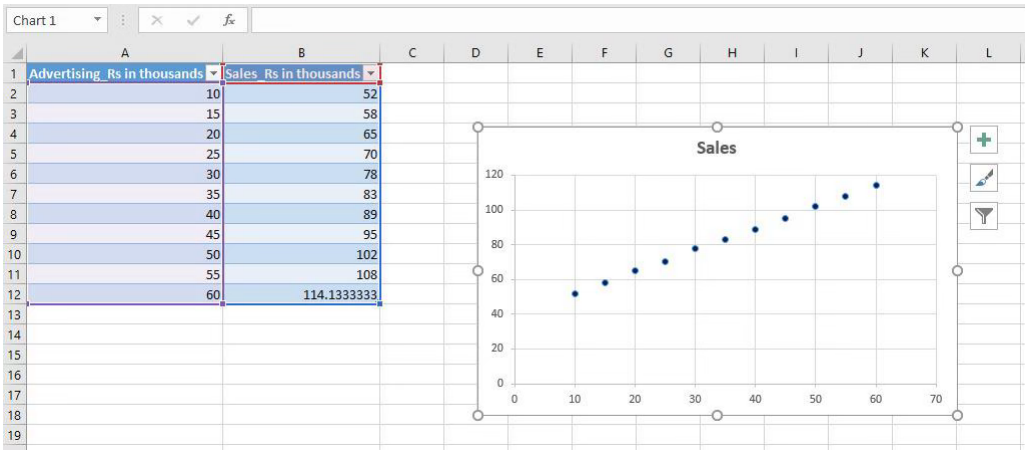
In Excel for Windows select File > Options > Add-ins. In Manage choose Excel Add-ins > Go. Tick Analysis ToolPak and select OK. After activation, Data Analysis appears on the Data tab.



	A	B	C	D
1	Advertising_Rs in thousands	Sales_Rs in thousands		
2	10	52		
3	15	58		
4	20	65		
5	25	70		
6	30	78		
7	35	83		
8	40	89		
9	45	95		
10	50	102		
11	55	108		
12	60	114.1333333		
13				
14				

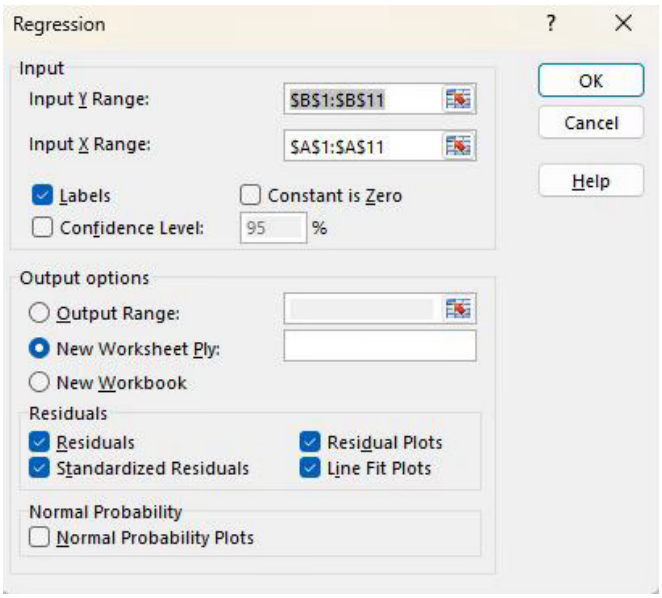
Step 2: Create a Scatter Plot

Select both columns. Choose Insert > Scatter > Scatter with only Markers. Put Advertising on X-axis and Sales on Y-axis.



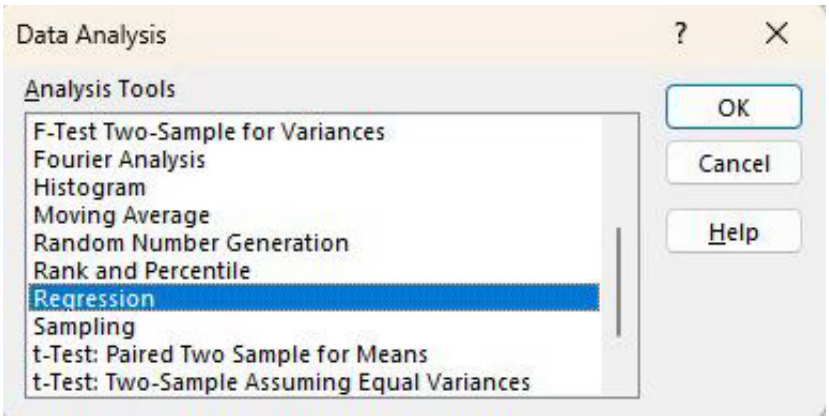
Step 3: Run Regression

Select Data > Data Analysis > Regression. Set Input Y Range to Sales including the header (B1:B11) and Input X Range to Advertising including the header (A1:A11). Tick Labels. Choose New Worksheet Ply for output and select OK.



Step 4: Interpret R Square

Locate R Square under Regression Statistics. A value close to 1 indicates that advertising explains a large proportion of variation in sales for this sample.

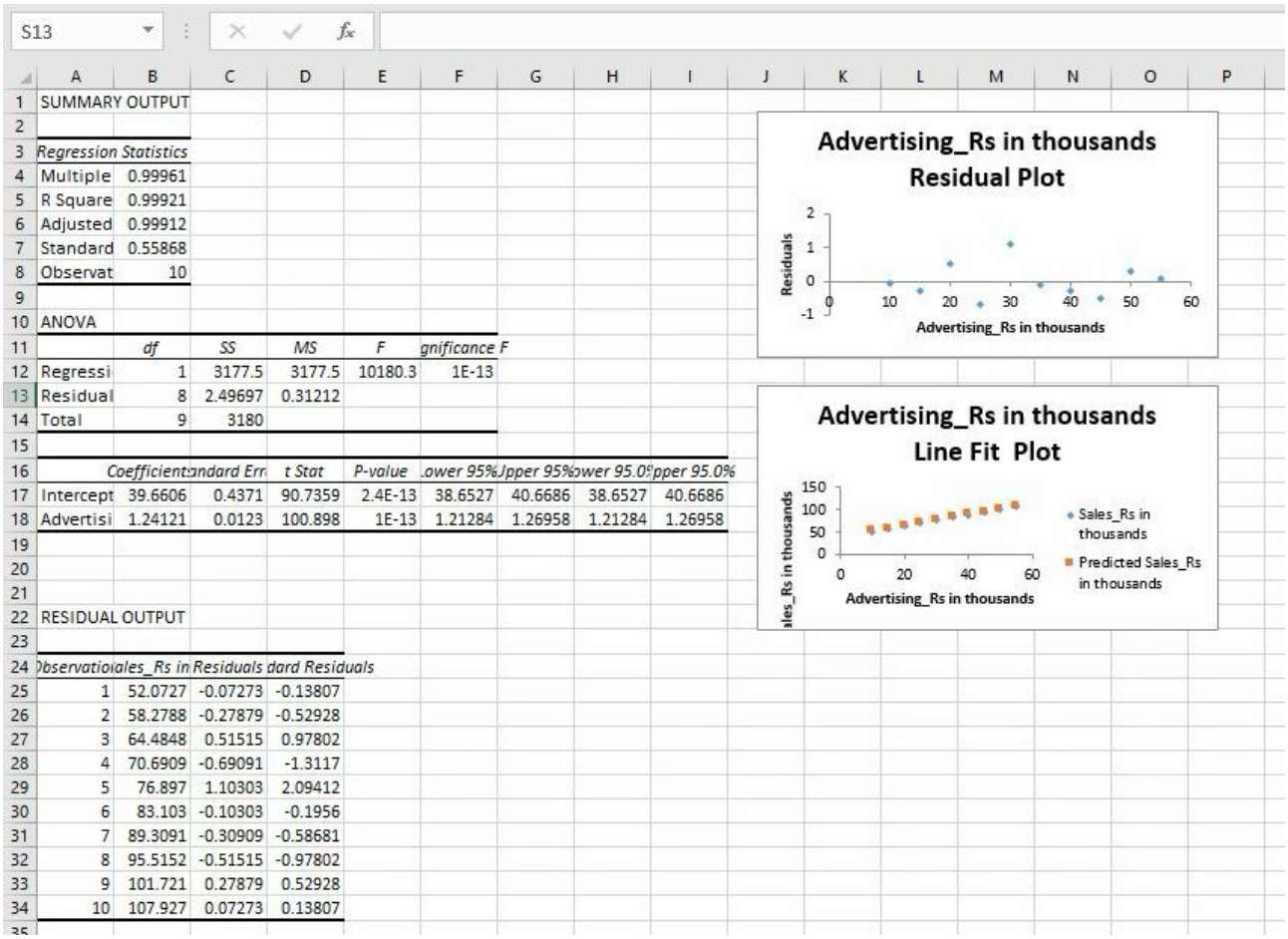


Step 5: Interpret Coefficients

In the Coefficients table, Intercept is a and X Variable 1 is slope b. The fitted equation is Sales = Intercept + Slope × Advertising.

Step 6: Interpret p-value

Check the p-value for X Variable 1. If p < 0.05, the advertising coefficient is statistically significant at the 5% level for this sample.



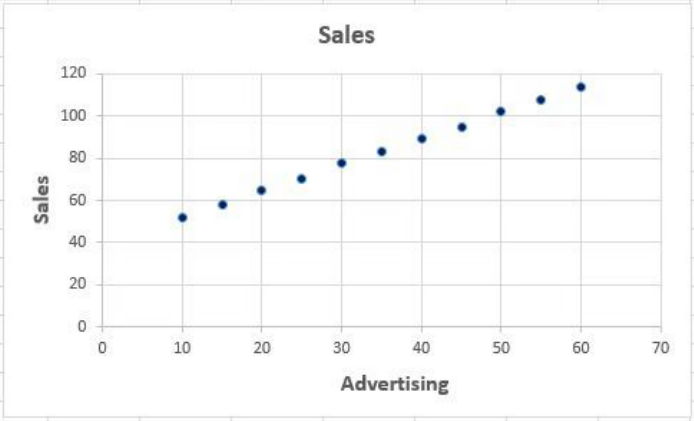
	A	B	C	D	E	F	G	H	I
1	SUMMARY OUTPUT								
2									
3	*Regression Statistics*								
4	Multiple	0.99961							
5	R Square	0.99921							
6	Adjusted	0.99912							
7	Standard	0.55868							
8	Observat	10							
9									
10	ANOVA								
11		*df*	*SS*	*MS*	*F*	*gnificance F*			
12	Regressi	1	3177.5	3177.5	10180.3	1E-13			
13	Residual	8	2.49697	0.31212					
14	Total	9	3180						
15									
16		*Coefficient:*	*ndard Err*	*t Stat*	*P-value*	*ower 95%*	*pper 95%*	*ower 95.0%*	*pper 95.0%*
17	Intercept	39.6606	0.4371	90.7359	2.4E-13	38.6527	40.6686	38.6527	40.6686
18	Advertisi	1.24121	0.0123	100.898	1E-13	1.21284	1.26958	1.21284	1.26958
19									
20									
21									
22	RESIDUAL OUTPUT								
23									
24	*Observatio*	*ales_Rs in*	*Residuals*	*dard Residuals*					
25	1	52.0727	-0.07273	-0.13807					
26	2	58.2788	-0.27879	-0.52928					
27	3	64.4848	0.51515	0.97802					
28	4	70.6909	-0.69091	-1.3117					
29	5	76.897	1.10303	2.09412					
30	6	83.103	-0.10303	-0.1956					
31	7	89.3091	-0.30909	-0.58681					
32	8	95.5152	-0.51515	-0.97802					
33	9	101.721	0.27879	0.52928					
34	10	107.927	0.07273	0.13807					

Step 7: Predict with FORECAST.LINEAR

To estimate sales when Advertising = 60, place 60 in A12 and enter the following formula in B12:

B12 =FORECAST.LINEAR(A12,$B$2:$B$11,$A$2:$A$11)

	A	B
1	Advertising_Rs in thousands	Sales_Rs in thousands
2	10	52
3	15	58
4	20	65
5	25	70
6	30	78
7	35	83
8	40	89
9	45	95
10	50	102
11	55	108
12	60	114.1333333



Result

A simple linear regression model was fitted and interpreted for advertising and sales data.

Conclusion

Regression supports business decisions by quantifying relationships and providing model-based predictions, while statistical significance and goodness-of-fit should also be considered.

Practical No. 8

Predictive Analytics using Simple Models (Trend and Forecast Analysis)

Aim : To analyse monthly sales trends and forecast future values using Excel.

Software / Tools Used

Microsoft Excel 2021 / 2024 / Microsoft 365 desktop (Forecast Sheet); FORECAST.LINEAR can be used as a compatible alternative

Objectives

- Create a time-series dataset.
- Visualize historical trend.
- Generate future forecasts using Forecast Sheet.
- Calculate a simple linear forecast using FORECAST.LINEAR.

Theory : Forecasting uses historical time-based data to estimate future values. Excel Forecast Sheet creates a forecast table and chart from a regular timeline. FORECAST.LINEAR provides a simple linear-regression-based prediction and is useful as a basic model for comparison.

Dataset

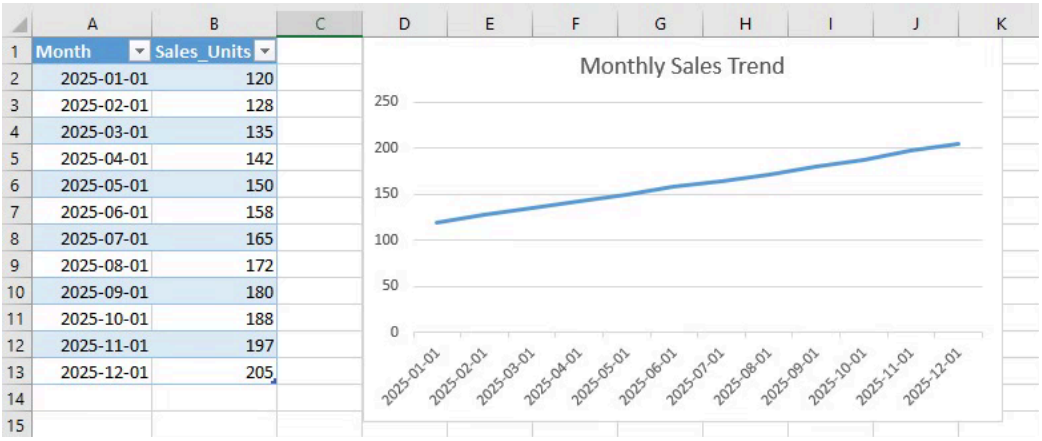
Month	Sales_Units
2025-01-01	120
2025-02-01	128
2025-03-01	135
2025-04-01	142
2025-05-01	150
2025-06-01	158
2025-07-01	165
2025-08-01	172
2025-09-01	180
2025-10-01	188
2025-11-01	197
2025-12-01	205

Step 1: Format Timeline

Enter the Month column as actual Excel dates and format it as mmm-yyyy. The timeline should use consistent monthly intervals.

Step 2: Plot Historical Trend

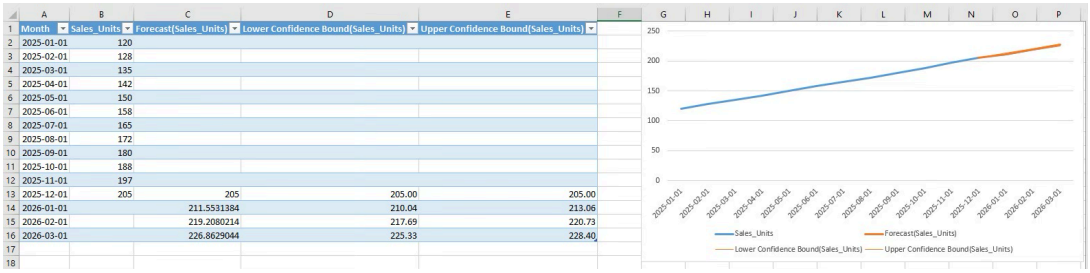
Select both columns and choose Insert > Line Chart > 2-D Line. Add the title Monthly Sales Trend.



Month	Sales_Units
2025-01-01	120
2025-02-01	128
2025-03-01	135
2025-04-01	142
2025-05-01	150
2025-06-01	158
2025-07-01	165
2025-08-01	172
2025-09-01	180
2025-10-01	188
2025-11-01	197
2025-12-01	205

Step 3: Create Forecast Sheet

Select both data series. Go to Data > Forecast Sheet. Choose the line-chart preview, set Forecast End to 1-Mar-2026, and select Create.



Month	Sales_Units	Forecast(Sales_Units)	Lower Confidence Bound(Sales_Units)	Upper Confidence Bound(Sales_Units)
2025-01-01	120			
2025-02-01	128			
2025-03-01	135			
2025-04-01	142			
2025-05-01	150			
2025-06-01	158			
2025-07-01	165			
2025-08-01	172			
2025-09-01	180			
2025-10-01	188			
2025-11-01	197			
2025-12-01	205	205	205.00	205.00
2026-01-01		211.5531384	210.04	213.06
2026-02-01		219.2080214	217.69	220.73
2026-03-01		226.8629044	225.33	228.40

Step 4: Simple Linear Forecast Alternative

Create Month_Number values 1 to 12 in a helper column. Enter 13, 14 and 15 for the next three months. Use FORECAST.LINEAR to predict each future Sales_Units value. If Month_Number is in C and Sales_Units in B, the formula for row 14 can be:

Month	Sales_Units	Helper	Forecast(Sales_Units)	Lower Confidence Bound(Sales_Units)	Upper Confidence Bound(Sales_Units)
2025-01-01	120	1			
2025-02-01	128	2			
2025-03-01	135	3			
2025-04-01	142	4			
2025-05-01	150	5			
2025-06-01	158	6			
2025-07-01	165	7			
2025-08-01	172	8			
2025-09-01	180	9			
2025-10-01	188	10			
2025-11-01	197	11			
2025-12-01	205	12	205	205.00	205.00
2026-01-01	211.4393939	13	211.5531384	210.04	213.06
2026-02-01	219.0967366	14	219.2080214	217.69	220.73
2026-03-01	226.7540793	15	226.8629044	225.33	228.40

Step 5: Compare and Interpret

Compare the Forecast Sheet results with the linear forecast. Explain that different forecasting methods can produce different values because they use different assumptions and treatment of trend/seasonality.

Result

Monthly sales were visualized and future values were forecast using Excel Forecast Sheet and a simple linear model.

Conclusion

Forecasting converts historical trends into future estimates, but predictions should be interpreted with awareness of model assumptions and uncertainty.

PRACTICAL NO 9
Customer Segmentation using K-Means Clustering

Aim : To segment customers into groups using K-Means clustering in Python.

Software / Tools Used
Google Colab; Python; pandas; matplotlib; scikit-learn

Objectives
- Create a customer dataset.
- Standardize numerical features.
- Apply K-Means with K = 3.
- Visualize and interpret customer clusters.

Theory : K-Means is an unsupervised learning algorithm that partitions observations into K clusters. Each observation is assigned to the nearest centroid, centroids are recalculated, and the process repeats until assignments stabilize. Standardization is useful when features have different scales.

Step 1: Create a New Colab Notebook
Open Google Colab and create a New Notebook. Run the following imports:

```
[1]
✓ 3s
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
```

Step 2: Create the Dataset (aka DataFrame)

```
[2]
✓ 0s
df = pd.DataFrame({
    'CustomerID': range(1, 13),
    'AnnualIncome': [15,16,17,18,55,58,60,62,85,88,90,92],
    'SpendingScore': [80,75,82,70,50,48,55,52,20,25,18,22]
})
print(df)
```

```
    CustomerID  AnnualIncome  SpendingScore
0            1            15             80
1            2            16             75
2            3            17             82
3            4            18             70
4            5            55             50
5            6            58             48
6            7            60             55
7            8            62             52
8            9            85             20
9           10            88             25
10          11            90             18
11          12            92             22
```

Step 3: Select and Standardize Features

```
X = df[['AnnualIncome', 'SpendingScore']]
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

Step 4: Apply K-Means

```
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
df['Cluster'] = kmeans.fit_predict(X_scaled)
print(df)
```

```
    CustomerID  AnnualIncome  SpendingScore  Cluster
0            1            15             80        2
1            2            16             75        2
2            3            17             82        2
3            4            18             70        2
4            5            55             50        0
5            6            58             48        0
6            7            60             55        0
7            8            62             52        0
8            9            85             20        1
9           10            88             25        1
10          11            90             18        1
11          12            92             22        1
```

Step 5: Examine Cluster Averages

```
print(df.groupby('Cluster')[['AnnualIncome','SpendingScore']].mean().round(2))
```

```
         AnnualIncome  SpendingScore
Cluster
0               58.75          51.25
1               88.75          21.25
2               16.50          76.75
```

Step 6: Visualize the Clusters

```
for c in sorted(df['Cluster'].unique()):
    part = df[df['Cluster'] == c]
    plt.scatter(part['AnnualIncome'], part['SpendingScore'], label=f'Cluster {c}')

plt.xlabel('Annual Income')
plt.ylabel('Spending Score')
plt.title('Customer Segmentation using K-Means')
plt.legend()
plt.show()
```

Step 7: Interpret

Use the cluster averages rather than the numeric label itself. In this dataset, one group has low income/high spending, one has medium income/medium spending, and one has high income/low spending.

Result

Customers were successfully segmented into three groups using K-Means clustering.

Conclusion

K-Means helps businesses discover homogeneous customer groups for differentiated marketing and service strategies.

PRACTICAL NO 10

Market Basket Analysis using Apriori Algorithm

Aim : To discover frequent itemsets and association rules from retail transactions using the Apriori algorithm.

Software / Tools Used

Google Colab; Python; pandas; mlxtend

Objectives

- Convert transactions to one-hot format.
- Find frequent itemsets using Apriori.
- Generate association rules.
- Interpret support, confidence and lift.

Theory : Market Basket Analysis discovers products that frequently occur together in transactions. Apriori first identifies frequent itemsets and then association rules can be generated. Support measures frequency, confidence measures conditional reliability, and lift compares co-occurrence against chance expectation.

Step 1: Install and Import Libraries

```
[1]
4s
!pip -q install mlxtend

import pandas as pd
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, association_rules
```

Step 2: Create Transaction Data

```
[8]
0s
transactions = [
    ['Milk','Bread','Butter'],
    ['Bread','Butter'],
    ['Milk','Bread'],
    ['Milk','Bread','Butter','Jam'],
    ['Bread','Jam'],
    ['Milk','Butter'],
    ['Milk','Bread','Butter'],
    ['Bread','Butter','Jam'],
    ['Milk','Bread','Jam'],
    ['Milk','Bread','Butter','Jam']
]
```

```
[9]
0s
te = TransactionEncoder()
te_array = te.fit(transactions).transform(transactions)
basket = pd.DataFrame(te_array, columns=te.columns_)
print(basket)
```

```
   Bread  Butter    Jam   Milk
0   True    True  False   True
1   True    True  False  False
2   True   False  False   True
3   True    True   True   True
4   True   False   True  False
5  False    True  False   True
6   True    True  False   True
7   True    True   True  False
8   True   False   True   True
9   True    True   True   True
```

Step 3: Find Frequent Itemsets

```
[10]
✓ 0s
frequent_itemsets = apriori(
    basket,
    min_support=0.30,
    use_colnames=True
)
print(frequent_itemsets.sort_values('support', ascending=False))
```

```
    support              itemsets
0       0.9               (Bread)
1       0.7              (Butter)
3       0.7                (Milk)
4       0.6        (Bread, Butter)
6       0.6          (Bread, Milk)
2       0.5                 (Jam)
5       0.5           (Jam, Bread)
8       0.5          (Milk, Butter)
11      0.4  (Bread, Butter, Milk)
7       0.3          (Jam, Butter)
9       0.3            (Jam, Milk)
10      0.3   (Jam, Bread, Butter)
12      0.3     (Jam, Bread, Milk)
```

Step 5: Generate Association Rules

```
[11]
✓ 0s
rules = association_rules(
    frequent_itemsets,
    metric='confidence',
    min_threshold=0.60
)

print(rules[['antecedents','consequents','support','confidence','lift']]
      .sort_values('lift', ascending=False))
```

```
         antecedents      consequents  support  confidence      lift
10    (Jam, Butter)          (Bread)      0.3    1.000000  1.111111
2             (Jam)          (Bread)      0.5    1.000000  1.111111
16      (Jam, Milk)          (Bread)      0.3    1.000000  1.111111
6             (Milk)         (Butter)      0.5    0.714286  1.020408
7           (Butter)           (Milk)      0.5    0.714286  1.020408
17            (Jam)    (Bread, Milk)      0.3    0.600000  1.000000
11            (Jam)  (Bread, Butter)      0.3    0.600000  1.000000
13    (Bread, Milk)         (Butter)      0.4    0.666667  0.952381
12  (Bread, Butter)           (Milk)      0.4    0.666667  0.952381
0           (Bread)         (Butter)      0.6    0.666667  0.952381
1           (Butter)          (Bread)      0.6    0.857143  0.952381
3           (Bread)           (Milk)      0.6    0.666667  0.952381
4             (Milk)          (Bread)      0.6    0.857143  0.952381
14   (Milk, Butter)          (Bread)      0.4    0.800000  0.888889
9      (Jam, Bread)         (Butter)      0.3    0.600000  0.857143
8             (Jam)           (Milk)      0.3    0.600000  0.857143
5             (Jam)         (Butter)      0.3    0.600000  0.857143
15     (Jam, Bread)           (Milk)      0.3    0.600000  0.857143
```

Step 6: Interpret a Rule

Choose a rule with lift greater than 1. Explain that the antecedent and consequent appear together more often than expected if they were independent.

Result

Frequent itemsets and association rules were generated from retail transactions using Apriori.

Conclusion

Apriori-based market basket analysis can support cross-selling, product bundling, recommendation and shelf-placement decisions.